

The Causal Effects of Korea's Quota Tariff Policy on Agricultural Product Retail Prices

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Abstract

The Korean government operates the Quota Tariff system with the objective of stabilizing prices. This study examines whether the policy causally lowered agricultural retail prices as intended. The sample is a daily panel of retail prices for 41 agricultural products from January 2021 through March 2025. Eleven of these products (six leafy and root vegetables and five fruits) experienced tariff reductions under the Quota Tariff; the remaining 30 did not. Quota Tariff designations begin at different times across products (staggered adoption), the treatment is suspended and later resumed (non-absorbing), and the depth of the tariff reduction also differs across products (heterogeneous treatment intensity). Since existing estimation methods that presuppose absorbing treatment cannot be applied to this structure, we adopt local projection difference-in-differences as the best-suited method. We then estimate two effects: the effect while the tariff reduction remains in force (the persistent-treatment specification) and the total effect of a Quota Tariff designation (the impulse-response specification). For the vegetable group, retail price declines were insignificant in both specifications, whereas the fruit group showed declines of about 0.78% per one percentage point of tariff reduction at 150 days after Quota Tariff onset (persistent treatment) and 0.49 ~ 0.55% at about 250~273 days (impulse response). The Quota Tariff entails large policy costs. Concentrating designations on fruits, rather than on vegetables for which no effect is identified, can therefore stabilize prices at a lower fiscal cost.

Keywords: Quota Tariff, Price Stabilization, Local Projection Difference-in-Differences, Agricultural Trade Policy

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1 Introduction

This paper studies Korea's Quota Tariff system, grounded in *Article 71 of the Customs Act*, focusing on the tariff reductions aimed at stabilizing agricultural prices. The Quota Tariff is a flexible tariff instrument: by Presidential Decree, the government can adjust both the reduced tariff rate and the volume of imports eligible for that rate. This allows the government to charge a rate below the Basic Tariff Rate, supporting the smooth supply of goods and promoting domestic price stability. Moreover, a quantity limit is not an essential element of the system, which allows a temporary tariff reduction to be granted without any volume restriction. This sets the Quota Tariff apart from the Tariff-rate Quota (TRQ) operated under the World Trade Organization (WTO) and free trade agreements (FTAs). The TRQ guarantees the market access a country has committed to under the WTO Agreement on Agriculture or an FTA, applying a low tariff rate up to a specified volume and a higher rate to imports beyond it, and so pursues market opening and the protection of domestic industry at the same time. Korea's Quota Tariff, by contrast, adjusts tariffs flexibly for domestic policy purposes independently of such international obligations, and is in this respect a far more flexible instrument than the TRQ. To keep the two distinct, this paper uses the term 'Quota Tariff' to refer to Korea's system.

Prior studies have focused mainly on the trade and protection effects of the TRQ (Loginova et al., 2021; Schaefer and Wolf, 2025), whereas empirical analyses of the Quota Tariff are scarce; this paper fills that gap by analyzing the causal effect of the Quota Tariff on consumer prices. The gap extends beyond the choice of subject to the methodology. The tariff measures imposed by the TRQ family and by the Quota Tariff are temporary, and they therefore generate a non-absorbing treatment: one that starts and ends repeatedly within the sample period. Yet studies that estimate the causal price effects of such institutions directly on non-absorbing data are, to the best of our knowledge, limited to a single one, Loginova et al. (2021). This study instead estimates the effects dynamically from the non-absorbing treatment path itself, using a daily treatment record that reflects even the lapses caused by quota exhaustion; section 2 reviews this methodological gap in detail.

Finally, the study also has important implications for Korea's Quota Tariff policy on agricultural products, where debate over the price-stabilizing effect of the Quota Tariff

has persisted for years. As of 2024, coverage of agricultural products had expanded considerably: the number of covered products rose from 20 in 2020 to 72 in 2024, and total tariff reductions reached about one billion dollars, a substantial fiscal cost (Jang, 2025). Since about 66% of these measures were deployed to stabilize retail prices, it is important to assess whether the policy actually achieves its original goal.

To answer this question, this paper constructs a daily panel of retail prices covering 41 agricultural products from January 2021 through March 2025: eleven products that had the Quota Tariff in force at least once during the sample period (six leafy and root vegetables and five fruits) and 30 products that never did. Designations begin on different dates across products (staggered adoption), a lapsed designation can be renewed on the same product (non-absorbing treatment), and the depth of the tariff reduction differs widely (heterogeneous treatment intensity). Conventional difference-in-differences (DiD) estimators are either biased under this structure or presuppose absorbing treatment. This paper therefore adopts local projection difference-in-differences (LP-DiD), which accommodates non-absorbing treatment (Dube et al., 2025). For each of the two treatment definitions, treatment status (binary) and treatment intensity (continuous), the estimation targets the effect of treatment while the tariff reduction remains in force (persistent treatment) and the total effect of a Quota Tariff designation (impulse response). Throughout, the analysis presupposes the Stable Unit Treatment Value Assumption (SUTVA): one product's outcome is unaffected by another product's treatment.

For the vegetable group, composed of leafy and root vegetables, the estimated retail price reductions are insignificant in every specification, whereas for fruits a significant and persistent reduction is found. The estimated response of fruit retail prices is about -0.78% per one percentage point (%p) of tariff reduction at 150 days after onset (the effect while the tariff reduction remains in force) and $-0.49 \sim -0.55\%$ at 250~273 days after onset (the total effect of initiation). Full and immediate pass-through would imply $0.77 \sim 0.94\%$ per 1%p; these magnitudes therefore indicate that a substantial share of the tariff reduction reached consumers, at the shorter horizon almost all of it. These results suggest that refraining from Quota Tariff designations on vegetables, for which no price-reducing effect is identified, would substantially reduce policy costs, and that the Quota Tariff is effective when concentrated on products with a high import share, such as fruits.

The data and code needed to replicate this study are publicly available.¹ The remainder of the paper proceeds as follows. Section 2 presents the institutional background and related literature, and section 3 describes the construction of the dataset. Section 4 estimates the causal effect of the Quota Tariff on retail prices using LP-DiD, section 5 examines the robustness of those estimates, and section 6 summarizes the findings and their policy implications.

2 Institutional Background and Related Literature

The TRQ originates in the Uruguay Round: the WTO Agreement on Agriculture converted non-tariff barriers into tariffs and, in exchange, introduced the TRQ to guarantee minimum market access. It has since become a principal instrument for opening agricultural markets not only under that agreement but also under FTAs. Tariffs, however, also serve domestic policy objectives. Institutional designs differ across countries, but temporary tariff adjustments to relieve supply shortages or stabilize prices can be found in a number of them. Korea's Quota Tariff, the subject of this study, is one such instrument, operated under *Article 71 of the Customs Act* separately from the TRQs of the WTO and FTA frameworks. Because it operates by Presidential Decree rather than through the legislative process of the National Assembly, the tariff rate and the volume of imports eligible for that rate can be adjusted quickly and flexibly as market conditions change.

The existing literature has mainly analyzed the economic effects of TRQs administered within the WTO and FTA frameworks. Loginova et al. (2021) analyze how Switzerland's administration of agricultural TRQs affects domestic price formation. Swiss annual quotas for agricultural products (excluding meat and dairy) are set above import demand, compensating for insufficient domestic production capacity while fulfilling the country's WTO market access quota obligations; during the seasons when domestic produce comes to market, however, the system is administered so as to protect domestic producers. The authors find that during these protection periods, Swiss vegetable prices were on average more than 20% higher than those in neighboring countries, with price gaps exceeding 50% for some products. These price effects, however, were not uniform across all agricultural products.

¹https://github.com/jayje0/github_TRQ

In addition, [Schaefer and Wolf \(2025\)](#) analyze the reform of Canada’s dairy TRQ system. A dispute-settlement panel under the United States–Mexico–Canada Agreement ruled that Canada’s procedures for allocating dairy TRQs were inconsistent with the market-access obligations of the agreement, and the revision that followed generated additional trade expansion for some dairy products. The literature has thus focused mainly on the market-access and trade effects of TRQs. By contrast, empirical evidence on the consumer-price effects of tariffs used as an instrument of domestic price stabilization remains insufficient. For Korea’s Quota Tariff, [Song \(2023\)](#) analyzes the relationship between the import price excluding tariffs and consumer prices, but this approach falls short of directly identifying the policy effect of the Quota Tariff itself. This study therefore addresses this limitation of the prior literature by causally identifying the consumer-price stabilization effect of the Quota Tariffs deployed to stabilize agricultural prices.

This gap has a methodological dimension as well, one that concerns the temporal structure of the treatment that institutions of this kind generate. Institutions in the TRQ family and the Quota Tariff differ in purpose and legal basis, but they share the feature that matters most for causal evaluation: the tariff measures they impose are temporary, starting and ending repeatedly within the sample period. Protection phases of seasonal TRQs open and close every year with the domestic harvest calendar ([Loginova et al., 2021](#); [Hillen, 2019](#)); safeguard tariffs under trade agreements are triggered when cumulative imports cross a threshold volume and are lifted afterwards ([Livingston et al., 2025](#)); and a Korean Quota Tariff designation likewise lapses when its notified end date arrives or its quota is exhausted, with the same product often designated again later.

The DiD literature calls a treatment absorbing when, once switched on, it remains on until the end of the sample, and non-absorbing when it can switch off and later switch back on ([Bach et al., 2025](#); [Dube et al., 2025](#)); the formal definitions appear in section 3. By that standard the treatments administered by the institutions above are all non-absorbing. Even so, empirical studies that estimate the causal price effects of these institutions directly on non-absorbing data are scarce: to the best of our knowledge, the [Loginova et al. \(2021\)](#) study reviewed above is the only one. Studies of causal price effects have, conversely, excluded this switching at the design stage.² The other

²For example, [Flaen et al. \(2020\)](#) estimate the retail-price effect of the 2018 U.S. washing-machine safeguard, a TRQ charging 20% within a 1.2-million-unit quota and 50% beyond it in its first year; because

studies that exploited this switching structure in estimation had outcome variables other than prices, such as market integration under the Swiss seasonal TRQs (Hillen, 2019) and bilateral trade flows under the beef safeguard of the United States–Japan trade agreement (Livingston et al., 2025). It is by estimating causal price effects directly on non-absorbing data that the present paper hopes to contribute.

Whether a study engages the non-absorbing path directly is not a technical detail: it determines which questions about this class of policies can be answered at all. For a temporary instrument the key questions are two: whether prices respond while the tariff reduction remains in force, and whether the response outlives the designation itself. These correspond exactly to the two estimands developed in section 4, persistent treatment and treatment initiation. An absorbing design that codes treatment as ‘ever designated’ counts quota-exhausted and expired days as treated. The resulting coefficient mixes two distinct periods—the days when the reduction was actually in force and the days after it had lapsed—so it can answer neither question: not whether prices fell while the policy operated, nor whether the fall outlasted it. Ignoring the switching structure also carries an econometric cost: observations still carrying the residual effects of a lapsed episode blend into the control side and contaminate the estimates, which is exactly what the *clean-control* and *washout* discipline of this study prevents. The contributions of this study therefore extend to (1) the institutional character of its subject, (2) the policy implications of its findings, and (3) the econometric methodology, a combination that no study of which the authors are aware has yet achieved.

3 Data Construction

The dataset constructed for this study is a product-level daily panel covering 41 agricultural products from January 1, 2021 through March 31, 2025. Daily retail prices,

their sample ends while the policy is still in place, the design treats its adoption as a single absorbing event. Fajgelbaum and Khandelwal (2026) apply the local projection event-study design of Dube et al. (2025), devised for non-absorbing treatments, to the 2025 U.S. tariffs, and yet state explicitly that they define treatment as “an absorbing state that remains on even if the applied rate falls in subsequent months.” Even Loginova et al. (2021), the sole exception, uses the switching only statically: it pools the weekly observations of the protected and unprotected phases across years, estimates no dynamics by horizon, and does not consider that the effects of a protection phase may persist into the following unprotected phase, which serves as the control. That design works because Swiss switching follows a regular seasonal calendar; Korea’s Quota Tariff switches irregularly with emergency designations and quota exhaustion, which leaves phase pooling unavailable here.

obtained from *Nongnet*, serve as the dependent variable of the causal analysis. The data on Quota Tariff status, designation start and end dates, and treatment intensity by product (the most critical explanatory variables in the regressions) were compiled from the public records published under the *Regulations on the Application of Quota Tariffs under Article 71 of the Customs Act*. This section describes how these variables, and the covariates for two important confounders, are constructed; the remaining covariates are described briefly where they enter the analysis in section 4.

Appendix A presents the summary statistics underlying the LP-DiD analysis: table 4 covers every variable entering the regressions, while tables 5 and 6 report the annual import volumes and values of the 41 products, showing which of them rely heavily on imports and which do so only marginally.

3.1 Selection of Treated and Controls

Since the analysis relies on LP-DiD, selecting the treated and control groups is essential. A structural feature of LP-DiD, however, is that a treated item can also serve as a control at some horizons (h), as section 4.2 discusses in detail. The selection is therefore stated more precisely in terms of an ever-treated and a never-treated group. An ever-treated item is one that had the Quota Tariff in force at least once during the sample period, and depending on the horizon it functions sometimes as treated and sometimes as a control; a never-treated item never had the Quota Tariff in force and functions only as a control.

The inclusion and exclusion rules deserve a full account, because it matters for the credibility of the design that group membership is determined by observable criteria, not by arbitrary selection on our part. The analysis is confined to agricultural products: livestock, fishery, and forest products lie outside its scope, which excludes foods such as chicken and eggs, as well as mushrooms, which official statistics group with agricultural products but which are legally classified as forest products.³ Items whose daily retail price series (the dependent variable) are unavailable or mostly missing are likewise excluded: broccoli, cherry, Korean melon, mustard greens, orange, peach, strawberry, and sweet persimmon. Potato is excluded for a different reason: its Quota Tariff applies only to potatoes for chip manufacturing, whereas retail prices

³Forestry and Mountain Villages Development Promotion Act.

are observed only for table potatoes; the treatment and the dependent variable would therefore refer to different goods. Grape was designated once in May 2024, but its daily retail price series is unobserved from June 2023 onward, leaving no dependent variable over its event window. Rice is also excluded: as the staple grain it is managed under the government’s priority food policy, under which its import and supply decisions differ from those of ordinary agricultural products.

Among the 41 products so selected, any item for which the Quota Tariff was in force at least once during the sample period (January 1, 2021 through March 31, 2025) is classified as ever-treated and the rest as never-treated. The eleven ever-treated items are listed by group. The leafy and root vegetables are napa cabbage, cabbage, radish, carrot, onion, and green onion; the fruits are mango, kiwifruit, banana, avocado, and pineapple. The remaining 30 products are all never-treated; panel C of table 4 in Appendix A lists them, divided into never-treated vegetables and other crops and never-treated fruits.

3.2 Quota Tariff Treatment Status for Ever-Treated Items

Table 1 reports, on a daily basis, when each ever-treated item was under the Quota Tariff. Border lines indicate monthly boundaries; white cells denote days on which the Quota Tariff was not in force (*not-treated status*), whereas black cells denote days on which it was in force (*treated status*). Data prior to 2022 are omitted for space, although daily treatment status was constructed for all products from 2015 onward; designations were exceedingly rare from 2015 through 2022 and increased markedly from 2024 onward. The treatment path used in the estimation is the status recorded in this table with a single adjustment: administrative year-end gaps of at most 30 days within a notice sequence (nine gaps) are bridged, as detailed in section 4.3.⁴ Gaps longer than 30 days are never bridged and are used exactly as recorded.

The remainder of this subsection details the criteria by which the black and white cells in table 1 were determined. This matters because the treatment status is the foun-

⁴The bridge is dictated by the design. Since the gaps run 13 to 23 days, far shorter than the 120-day washout, treating each renewal as a new onset would make every such re-onset dirty by construction and discard it from the treated side; in the persistent-treatment specifications the stay-on condition would in addition fail on the first day of the gap. A single continuous policy episode would thus be split into a short event plus an unusable orphan segment, distorting the object of measurement, namely the effect of treatment while it remains in force. Section 4.2.2 details the washout and stay-on conditions.

Table 1: Quota Tariff Treatment Status

[illegible]

dation of the LP-DiD analysis: it is what identifies the clean controls and clean treated units at the core of the design, and it must therefore be constructed accurately from the factual record. As the table shows, the treatment status in this study is ‘non-absorbing.’ A configuration in which white cells appear first chronologically and, once transitioning to black, never revert to white is termed ‘absorbing’ (Bach et al., 2025); configurations permitting reversion from black to white are termed ‘non-absorbing’ (Dube et al., 2025), which allows for indefinite white-black-white-black repetition.

A first distinction is between designations announced with a quota limit and those announced without one. Under a limited designation the tariff relief is available only until cumulative imports, counted from the start of the designation, reach the quota; once the quota is exhausted, the relief lapses even though the announced period has not ended, and the item must be regarded as reverting to not-treated status. Two notions of duration must therefore be kept apart: the *designated period*, announced in advance in the government notice, and the days on which the Quota Tariff is actually *in force*, which is what the treated status in table 1 records. Under an unlimited designation the two coincide and the relief runs uninterrupted to the end of the designated period.

Determining the exhaustion date requires care, because the cumulative imports counted against the quota are not the item's total imports but only the imports cleared through customs under the Quota Tariff. The Korea Customs Service discloses this

narrower series by item, but at monthly rather than daily frequency, which reveals the month in which the quota was exceeded but not the exact day; apart from this limitation it is the most authoritative source available (Method A). Alternatively, the exact crossing day can be computed from the item’s monthly total import volume under the assumption that each month’s imports are distributed uniformly across its days (Method B). The drawback is that Method B rests on total imports rather than only those cleared under the Quota Tariff, and it therefore overstates the relevant volume. This study combines the two methods to arrive at the most defensible exhaustion date for each episode.

Pineapple illustrates the procedure. Of the six Quota Tariff designations announced for it during the sample period, only one ended in exhaustion: under the designation running from August 25 to December 31, 2023, the quota was crossed on October 9, 2023, making October 8 the last day on which the Quota Tariff was in force, and the cells from October 9 onward within the designated period are white in table 1 (not-treated status). In the other five cases the quota never bound. Table 7 in Appendix B sets out the announced period and the quota of each of the six designations, and that appendix works through every case.

3.3 Anticipation of the Quota Tariff Policy

A further feature of the treatment path bears on identification: how far agents could see a designation coming. Full anticipation would lead importers and distributors to defer imports until the lower tariff took effect and to run down inventories ahead of the expected price decline, and such behavior would affect the estimates. For the agricultural Quota Tariff, however, the scope for anticipation is limited. Unlike industrial raw materials and similar items, for which the government announces an annual Quota Tariff operation plan in advance, agricultural designations are mostly emergency responses to current market conditions. Some measures took effect with no advance legislative notice at all; where notice was given, a lead of about one month occurred in only a few cases and for most measures the interval was shorter still, and once the decree was promulgated, enforcement typically began immediately or within one to five days. The policy thus leaves economic agents no practical window in which to adjust before it takes effect. The analysis accordingly does not treat the legislative

notice as a separate policy shock, and identifies the policy effect from the date each designation takes effect.

3.4 Treatment Intensity

The LP-DiD analysis is conducted for two treatment definitions. Treatment status is the effect on consumer prices of a Quota Tariff being in force, captured in the regressions by a dummy variable for treated status; treatment intensity is the effect on consumer prices of each 1%p of tariff reduction generated by a designation, captured by a continuous variable measuring how far the designation lowers the tariff rate relative to its pre-designation level. Computing the latter requires two pieces of information: the tariff rate charged while the Quota Tariff is in force, which is read directly from the government notice, and the tariff rate that would prevail in its absence, a rate we denote the ‘applied tariff rate.’

Deriving the applied tariff rate requires complex calculations across tariff agreements and Basic Tariff Rates for each country, product, and month, a task demanding specialized expertise in customs administration. Multiple rates coexist for each Harmonized Tariff Schedule of Korea (HSK)⁵ code, including Basic Tariff Rates and Preferential Tariffs under an FTA, and *Article 50 of the Customs Act* governs the priority order in which they apply. Even for identical products, the rate actually applied may vary with the country of origin, product specifications, and importer circumstances. No single statutory rate can therefore represent the tariff burden on a product, and an indicator such as the applied tariff rate is needed.

3.4.1 Derivation of the Applied Tariff Rate

This indicator is constructed from actual import records. Country-specific monthly import volumes by product⁶ drawn from the *Korea Agricultural Trade Information* (KATI) serve as the weights, since the rate that applies to the same product differs with the exporting country. For each country and month, the ‘most favorable tariff

⁵The HSK extends the internationally harmonized six-digit subheadings of the Harmonized System to ten-digit national tariff lines, at which the Korean government defines individual products and assigns tariff rates. The four digits beyond the sixth are specific to Korea and do not correspond to the tariff lines of other countries.

⁶Ten-digit HSK code.

rate’ applicable to that country’s imports was determined by following the ‘Tariff Rate Priority Structure’ established under the *Customs Act* and presented in Appendix C. The monthly applied tariff rate is the average of these country-specific rates weighted by import volume; repeating the calculation for every month from January 2021 through March 2025 yields a product-level panel of monthly applied tariff rates. This procedure cannot verify directly whether Preferential Tariffs were applied in individual transactions, but by combining the structure of the tariff system with observed import patterns it provides a meaningful estimate that closely approximates the tariff burden actually incurred.⁷

Two worked examples show what this produces. For banana the volume-weighted applied tariff rate was approximately 23.3% in July 2024, and fell to approximately 14~15% in the first quarter of 2025 after the Korea-Philippines FTA entered into force. For fresh unpeeled onion the rate swings between 50% and 135% from month to month: Chinese fresh unpeeled onion, the bulk of import volume, derives no practical benefit from the FTA Tariff Concessions, leaving the WTO Tariff Concession to apply, at the Recommended Tariff Rate within the Market Access Quota and at the non-recommended rate beyond it. Appendix D traces both examples country by country and plots the resulting monthly series of applied tariff rates in figure 7.

3.5 Removing Seasonality

This paper uses deseasonalized prices as the dependent variable, a choice justified as follows. Seasonality is removed through a regression-based adjustment using calendar dummy variables and Fourier terms (Pierce et al., 1984), designed to preserve the treatment effects. For daily series, standard official seasonal-adjustment tools (e.g., X-12/X-13-ARIMA-SEATS) are not directly applicable because they are designed for monthly or quarterly data and assume fixed 12-month or 4-quarter cycles (Mollins and Lumb, 2024). The natural alternative, Seasonal-Trend decomposition based on Loess (STL), is a non-parametric method that has been applied extensively to weekly and daily data (Cleveland et al., 1990); because it relies on non-parametric smoothing, however, part

⁷A further limitation concerns items for which the law prescribes charging whichever of the ad valorem duty and the specific duty is larger in amount: because the price and weight of individual import transactions are unobservable, the duty actually charged cannot be determined in such cases, and the applied tariff rate is therefore computed from ad valorem rates only.

of a trend or policy shock may be absorbed during the decomposition, and treatment effects removed at the seasonal-adjustment stage would distort the subsequent DiD analysis. The regression-based adjustment with calendar dummies and Fourier terms is chosen precisely because it removes the seasonal repetition while leaving the policy effect intact.

One might nevertheless object that, because the Quota Tariff is deployed precisely to suppress prices when they surge seasonally, the analysis should rely on the original prices. This objection can be countered on two grounds. (a) The pure effect of the Quota Tariff survives deseasonalization: the timing of Quota Tariff designations in Korea is far from a regular seasonal pattern, as table 1 shows, and the adjustment removes only recurring patterns, not genuine shocks, so that effect becomes even more transparent once the seasonal repetition is stripped away. (b) Using the original prices within the LP-DiD framework would obscure that effect, because each item has its own seasonal frequency and amplitude, making treated and control items not directly comparable.

The detailed implementation of the deseasonalization is documented in Appendix E, where figure 8 illustrates the procedure with the original and seasonally adjusted series of spinach retail prices.

3.6 Confounding Effects

The Quota Tariff is not the only instrument the government deploys against agricultural price surges. Two further instruments share both its trigger (supply shortfalls and surging prices) and its purpose of stabilizing consumer prices: temporary expansions of WTO Market Access Quotas and releases of government-stockpiled produce. Either can move retail prices at the same time as a Quota Tariff designation, and leaving them uncontrolled risks conflating their effects with the effect under study. The estimation therefore includes covariates for both, collected in the blocks Q_{it} and R_{it} of the covariate vector in section 4.3.

The first instrument operates within the conventional WTO-style TRQ, from which section 2 distinguishes the Quota Tariff: imports within a scheduled Market Access Quota enter at a low in-quota rate, while imports beyond it face a substantially higher over-quota rate. Only the items for which Korea’s WTO concessions schedule a quota (onion, garlic, and ginger among them, but not banana, avocado, or napa cabbage) can

experience an expansion, which the government may grant temporarily when supply–demand imbalances intensify, as it has done for sesame and onion. Since an expansion admits a larger import volume at the low in-quota rate, it is a tariff-relief measure deployed on the same grounds as the Quota Tariff. The expansion history is compiled from the enactment records and coded by a simple rule: the quota runs by calendar year, with cumulative imports counted from January 1st, under which an expansion takes effect on its enactment date and remains in force through December 31st, and a later expansion in the same year supersedes it from its own effective date.⁸

The block Q_{it} accordingly contains three covariates: a coverage dummy for items under a WTO TRQ; the expansion ratio in force on day t (the additional volume granted by the active expansion relative to the scheduled quota, equal to zero when no expansion is in force); and the expansion ratio interacted with the tariff gap between the over- and in-quota rates, which scales each expansion by the size of the tariff gap it relieves.⁹ The coverage dummy is there because the two kinds of zero are not equivalent: for an item outside the WTO TRQ system a zero expansion is forced, there being no quota to enlarge, whereas for a covered item it records the government’s choice not to deploy an available instrument.

The second instrument is the release of government stockpiles: the government stores staple produce in normal times and releases it when supply tightens or prices surge, with napa cabbage, radish, and dried red pepper among the prominent cases. A release presupposes a stockpile, and a recorded zero therefore means different things for different products: for most items no stockpile program exists and the release value is necessarily zero, whereas for a stockpiled item a zero records a month in which the program chose not to release. The release records are monthly totals in tons; mapped onto the daily panel, each monthly total applies to every day of its month, and products absent from the records are coded zero. The block R_{it} contains two covariates built from these monthly volumes: the inverse hyperbolic sine of the average release over the trailing window $[t - 130, t - 31]$, which ends before the month containing day t , and an incidence dummy equal to one when any release falls in that window. The inverse hyperbolic sine compresses the heavy right tail of the volume distribution while

⁸The one exception is ginger, whose 2023 expansion lapsed on 30 September 2023 by its own terms.

⁹All three are defined on every product-day, zero rather than missing, leaving no observation to drop from the regressions.

keeping the many zero values in the sample (Bellemare and Wichman, 2020), and the incidence dummy separates whether releases were under way from how large they were. Closing the window before the current month ($[t-130, t-31]$) is meant to ensure that only release information from before the Quota Tariff treatment date is taken into account; because the records are monthly, release information would otherwise spill over past the Quota Tariff treatment date and distort the estimated effect.

How far these two instruments, Q_{it} and R_{it} , overlap the Quota Tariff in practice can be read off the daily panel of the eleven ever-treated products (January 2021 through March 2025). The quota-expansion channel barely touches the treated set: onion is the only treated product covered by a WTO TRQ, and its single expansion episode (July~December 2023) does not intersect its Quota Tariff designation (August 2022~February 2023), leaving no product-day in the sample that carries both instruments at once. The expansions of the period fall instead on never-treated items (garlic, bean, mung bean, sesame, peanut, and ginger), whose prices enter the comparison group, and the Q_{it} covariates absorb those movements where they occur.

The stockpile channel, by contrast, overlaps the treatment substantially: the three stockpiled products among the eleven (onion, napa cabbage, and radish) all began their longest designation episodes in months with active releases (3,527, 955, and 1,959 tons, respectively), and releases were under way in the same month on 70% (573 of 815) of their days under the Quota Tariff. Within-product correlations between daily Quota Tariff status and the two release covariates are positive though moderate, 0.11~0.16 over the eleven products and 0.23~0.33 within the stockpiled three. The two instruments evidently respond to the same market conditions, which is exactly why the release covariates belong in the regressions: omitting them would risk crediting the Quota Tariff with the stabilizing effect of the releases. Section 4.4 examines how prices themselves move around designations, and how much of that movement the covariates absorb, once the estimating equations are in place.

4 Causal Analysis

This section estimates the causal effect of the Quota Tariff on retail prices in six steps. Section 4.1 explains why the treatment structure of section 3 calls for LP-DiD rather than conventional event-study estimators. Section 4.2 sets out the two estimands the

design delivers and the sample restrictions on which both rest, and section 4.3 turns them into estimating equations. Section 4.4 then probes the conditional parallel-trends assumption before any estimate is shown. Section 4.5 reports the estimates as dynamic response graphs, and section 4.6 condenses them into a regression table at selected horizons.

4.1 Why Local Projection Difference-in-Differences

Three features of the Quota Tariff, established in section 3, jointly determine the choice of estimator. Adoption is staggered: products enter treatment at different dates. Treatment is non-absorbing in the sense set out in section 3.2: a designation lapses at its notified end date or, for quota-limited notices, once the quota is exhausted, and the same product may be designated again later. Finally, treatment intensity is heterogeneous: the implied tariff reduction ranges from 1.5%p to 69.9%p across products (section 3.4). The analysis accordingly asks two questions on the same panel: whether Quota Tariff adoption as such lowers retail prices (a binary treatment-status question), and by how much retail prices fall per 1%p of tariff reduction (a continuous treatment-intensity question).

Conventional two-way fixed effects (TWFE) event-study regressions are ill-suited to this structure. When treatment turns on and off and effects are heterogeneous, they implicitly use previously treated observations as comparisons even though the lagged effects of an earlier episode may still be unfolding. The resulting contamination can severely bias the estimates, including through negative weighting (Goodman-Bacon, 2021; De Chaisemartin and D’Haultfœuille, 2020). Retail prices of pineapple in early 2024 are a poor counterfactual for anything, because its 2023 episode may persist well after quota exhaustion ended that episode.

The recent heterogeneity-robust estimators confront this contamination by restricting the controls to observations that are not yet treated or never treated (Roth et al., 2023; de Chaisemartin and D’Haultfœuille, 2023). But the interaction-weighted estimator of Sun and Abraham (2021) is constructed for the absorbing case, characterizing each unit by a single initial-treatment date, as does the irreversibility-of-treatment assumption of Callaway and Sant’Anna (2021). In contrast, the Quota Tariff is non-absorbing in *both* the treatment-status and the treatment-intensity specifications. Re-

coding treatment as an ever-treated indicator would change the estimand to the effect of *having ever* been designated and misclassify terminated and quota-exhausted periods as treated; relabeling each onset as a fresh absorbing cohort fails too, because several episodes adjoin so closely that any inter-episode “pre-period” would be contaminated by the preceding one.¹⁰

This study therefore employs LP-DiD (Dube et al., 2025), which is constructed for non-absorbing treatments: each horizon-specific regression compares newly treated observations only with contemporaneously¹¹ clean comparisons. Three further properties make it the natural choice here. It conditions directly on the pre-determined time-varying covariates, which the specification below requires. It nests the binary and the intensity treatments in one framework, so that differences between the two sets of estimates reflect the treatment definition rather than a change of estimator family. And it extends naturally to a treatment that is, as here, simultaneously non-absorbing and continuous: Dube et al. (2025) flag continuous treatment as a natural extension of their binary non-absorbing treatment, which this study implements through a pre-determined intensity scalar (see Dube and Lindner, 2024 for an LP-DiD application with non-absorbing treatment). The credibility of LP-DiD rests not on the regression equation but on the discipline it imposes: strict rules about which observations may serve as comparisons.

4.2 Core Design Elements of Local Projection Difference-in-Differences

The LP-DiD design has two components: a choice of estimand, and a pair of sample-selection rules — clean controls and clean treated units — governing which observations enter the horizon- h regression.

4.2.1 Two Estimands: Persistent Treatment versus Treatment Initiation

The *persistent-treatment* scheme asks what happens to prices while a designation is in force: it keeps a treatment event in the horizon- h regression only while the treatment

¹⁰The improved estimators for staggered designs proposed by Sun and Abraham (2021), Callaway and Sant’Anna (2021), Borusyak et al. (2024), and Gardner (2022) all presume that treatment, once switched on, stays on.

¹¹At the same horizon.

persists. The *treatment initiation* scheme asks what the total effect of having initiated a designation is, including whatever happens after the episode ends or the product re-enters; it follows every initiation across all horizons, leaving the post-initiation path as it actually unfolded. The coefficient path is thus an impulse response function (IRF): the average effect of treatment *initiation* h days later. This is the impulse-response-type estimand discussed by Dube et al. (2025), corresponding to the actual-versus-status-quo event-study effects formalized by De Chaisemartin and D’Haultfoeulle (2026). Neither estimand is more correct than the other; Dube et al. (2025) regard the choice as a matter of the research question, and here the two are complements. The persistent-treatment estimates speak to whether the policy lowers prices while it operates, the impulse-response estimates to whether initiation effects outlive the designations themselves, a question of first-order policy interest for a temporary instrument. The two schemes differ in exactly one sample condition on the treated side.

4.2.2 Clean Controls and Clean Treated

For a regression at horizon h , the comparison group at date t consists of product-day observations that have never been treated up to t and remain untreated through $t + h$. This “clean control” design, named in the stacked event-study literature on minimum wages (Cengiz et al., 2019) and formalized as a flexible clean control condition by Dube et al. (2025), follows from the contamination problem above. Units whose own treatment dynamics are still unfolding cannot reveal the untreated counterfactual. The same comparison group serves both schemes, *persistent-treatment* and *treatment initiation*, and the implementation here is deliberately stricter than the theory requires. The window-based condition of Dube et al. (2025) would readmit a previously treated product once its dynamics have stabilized; this study instead removes a product from the control pool permanently after its first treatment, which leaves an ever-treated product contributing clean control observations only before its first onset. Since 30 of the 41 products are never treated, clean controls remain abundant at every horizon.

A parallel discipline applies to the treated side, where the two schemes, *persistent-treatment* and *treatment initiation*, diverge. The *washout* condition requires the product to have been untreated throughout the 120 days preceding the onset, that is, over $[t - 120, t - 1]$. This is what makes an onset *clean*: because no treatment precedes it within the window, the dynamics attributed to the event reflect that event alone rather

than the residue of a recent previous episode. The condition implements the assumption of Dube et al. (2025) that the effects of a past episode stabilize within L periods, where $L = 120$ in this study. Since cleanliness concerns the past, the washout applies to every specification, the impulse-response ones included. In contrast, the *stay-on* condition is not about cleanliness. It retains a treated event at horizon h only if the product is treated on every day of $[t, t + h]$, which is what defines the persistent-treatment estimand, the effect of treatment still in force at $t + h$. It therefore applies to the persistent-treatment specifications only, and the impulse-response ones drop exactly this condition and nothing else. Together they reproduce the treated-side restriction that Dube et al. (2025) impose for non-absorbing treatments, and the same design logic runs through the stacked event-study literature (Cengiz et al., 2019; Wing et al., 2024; Dube and Lindner, 2024).

The treated-side and control-side are thus disciplined asymmetrically: a treated event needs a clean *recent history*, whereas a control observation needs a clean *entire history*. Pineapple shows that the washout is no formality: its onset of January 19, 2024 fails the condition, falling only 102 days after the preceding episode ended with quota exhaustion on October 8, 2023 (section 3.2). Nor can any pineapple observation after its first designation serve as a control, since the product’s treatment history is no longer clean. Appendix F classifies every onset—three fail the washout in all—and each classification follows mechanically without discretion. The treated-side conditions are used exactly as prescribed by Dube et al. (2025) and as implemented in the `lpdid` package for *STATA* and *R* (Busch and Girardi, 2024); only the control-side condition is stricter.

4.2.3 Horizon Coverage and Limits of the Two Estimands

The stay-on condition has a mechanical consequence that shapes how far the persistent-treatment estimates can be read. An event whose treatment spell lasts s days satisfies the stay-on condition only for horizons $h < s$, which drops shorter-spell events out one by one as h grows. Among the eleven ever-treated products, nine vegetable and eight fruit events survive the washout screen at $h = 0$. The vegetable set shrinks step by step at $h = 61, 104, 120, 156, 166$, and 177 , the fruit set at $h = 45, 52$, and 271 (Appendix F). Estimates at distant horizons therefore compare a different collection of events, a “composition effect” inherent to horizon-specific estimation (Dube et al., 2025). At

exactly $h = 156$, the day the cabbage event’s spell ends, the vegetable coefficient in the treatment-status specification jumps from +7.9 to +23.7 log points and turns statistically significant on impact, while the surviving treated set comes to be dominated by the radish and napa cabbage events of 2024, whose post-treatment windows coincide with the severe 2024 supply disruptions in those markets. A discontinuity that arrives on the day of a compositional break, rather than growing out of the preceding dynamics, is a composition artifact, not a treatment dynamic. The persistent-treatment graphs are therefore truncated at $h = 150$, a conservative margin below the break at 156, with the underlying estimates for all horizons up to 273 retained in the replication files. Attrition, however, is not a defect of the design but a direct consequence of the stay-on condition, which is precisely what defines the persistent-treatment estimand. The impulse-response specifications, having no stay-on condition, are free of attrition altogether: their treated counts stay at nine and eight across all horizons, and the impulse graphs extend to $h = 273$, the maximum the sample supports. The 2022 pineapple treatment event, for instance, enters the persistent-treatment regressions only through $h = 51$, where its tariff cut ends, but the impulse-response regressions through $h = 273$.

The impulse estimand also has a limitation that should be stated plainly. Treatment durations differ widely across events (spell lengths range from 45 to 585 days, the longest right-censored by the end of the sample), and the impulse coefficient averages over initiations without adjusting for the number of treated days each event accumulates within the window. It is therefore a per-initiation effect, not a per-treated-day effect. The same convention governs the intertemporal framework of [De Chaisemartin and D’Haultfœuille \(2026\)](#), whose event-study effects are likewise defined by the change in treatment alone, with no duration term entering the estimation. Read this way, the impulse coefficient answers the question a policymaker faces at the moment of designation: what initiating a Quota Tariff sets in motion for retail prices h days ahead.

4.3 Regression Specification

Following the pre-mean-differenced (PMD) LP-DiD of [Dube et al. \(2025\)](#), the dependent variable at horizon h is the log retail price h days after the event, measured against the

average log price over the 120 days preceding the event:

$$\bar{y}_{i,t-1} \equiv \frac{1}{120} \sum_{s=1}^{120} \ln P_{i,t-s}, \quad \tilde{\Delta}^h \ln P_{it} \equiv \ln P_{i,t+h} - \bar{y}_{i,t-1}, \quad -500 \leq h \leq 273, \quad (1)$$

where i and t denote product and calendar day, and P_{it} is the seasonally adjusted retail price (KRW/kg) of section 3.5. The pre-mean baseline is fixed before the event and is not an ad hoc choice: Dube et al. (2025) develop this PMD LP-DiD on the efficiency ground of Borusyak et al. (2024), that the pre-event level is estimated more precisely from many pre-treatment days than from the single day $t - 1$. The `lpdid` package in *STATA* and *R* implements it as a built-in option (Busch and Girardi, 2024), and this study adopts it unchanged. The argument has particular force in daily retail prices, where a single-day baseline would transmit that day’s idiosyncratic movement into the outcome at every horizon.¹² Since equation (1) differences each observation against its own pre-event level, product fixed effects are differenced out by construction.

For each horizon h , the treatment-status (binary) specification is the cross-sectional regression.¹³

$$\tilde{\Delta}^h \ln P_{it} = \beta^h (D_i \cdot \Delta D_{it}) + T_t + X_{it} + \varepsilon_{it}^h, \quad (2)$$

estimated, for $h \geq 0$, on the sample

$$\begin{aligned} \mathcal{S}_{\text{base}}^h = & \underbrace{\left\{ (i, t) : \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120], D_{i,t+j} = 1 \forall j \in [0, h] \right\}}_{\text{clean treated: fresh entry, remaining treated through } t+h} \\ & \cup \underbrace{\left\{ (i, t) : D_{is} = 0 \forall s \leq t, D_{i,t+j} = 0 \forall j \in [1, h] \right\}}_{\text{clean control: no history, untreated through } t+h}. \end{aligned} \quad (3)$$

Here $D_{it} \in \{0, 1\}$ is the daily Quota Tariff treatment status of table 1 and D_i indicates ever-treated products. T_t are calendar-day fixed effects, so that every event-time comparison is simultaneously a same-calendar-date comparison, absorbing market-wide

¹²The 120-day window matches the washout length and was fixed ex ante, in line with the recommendation of Dube et al. (2025) that the reference period be chosen on the characteristics of the application, independently of the estimation results. The size of the efficiency gain depends on the serial correlation of untreated outcomes (Borusyak et al., 2024; Dube et al., 2025). PMD does not by itself guarantee parallel trends; the pre-event coefficients reported in section 4.5 remain the relevant diagnostic.

¹³Although the data used in this study form a panel, LP-DiD selects every observation satisfying the clean-control and clean-treated conditions and then estimates the equation on that sample as a cross-sectional rather than a panel regression.

movements. The two sets in equation (3) are built separately rather than by matching treated to control observations one-to-one, and their dates t need not coincide: the horizon- h regression pools both into one sample, and the same-calendar-date comparisons are carried out not by the sample definition but by the calendar-day fixed effects. For $h < 0$ the dependent variable is the corresponding lag, and the forward conditions in equation (3) are dropped, namely the stay-on condition $D_{i,t+j} = 1 \forall j \in [0, h]$ on the treated side¹⁴ and the future no-treatment condition $D_{i,t+j} = 0 \forall j \in [1, h]$ on the control side:

$$\begin{aligned} \mathcal{S}_{\text{base}}^h = & \underbrace{\left\{ (i, t) : \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120] \right\}}_{\text{clean treated: fresh entry, washout only}} \\ & \cup \underbrace{\left\{ (i, t) : D_{is} = 0 \forall s \leq t \right\}}_{\text{clean control: no history through } t}, \quad h < 0. \end{aligned} \quad (4)$$

The washout restricts only the 120 days immediately preceding onset, not the treatment history before that window; at horizons $h < -120$ an event is therefore not excluded merely because $t+h$ falls within an earlier episode of the same product. On the control side, the no-history condition $D_{is} = 0 \forall s \leq t$ already covers every date up to t , leaving nothing further to impose at horizons $h < 0$.

The key regressor ΔD_{it} equals one only on the day product i enters treatment ($D_{i,t-1} = 0$ & $D_{it} = 1$). It is an onset indicator, not a treatment level: the regression compares the price path following a fresh designation with the contemporaneous paths of clean controls, while duration of treatment is handled by the sample restriction, not by the regressor. For $h \geq 0$, β^h therefore measures the average log price gap h days after a clean onset among events still under treatment; for $h < 0$ it is the pre-onset price gap underlying the pre-trend diagnostics of section 4.4 below.

The treatment-intensity (continuous) specification scales the onset indicator by the product-specific, pre-determined intensity G_i , in %p:

$$\tilde{\Delta}^h \ln P_{it} = \beta^h \left(G_i \cdot D_i \cdot \Delta D_{it} \right) + T_t + X_{it} + \varepsilon_{it}^h, \quad \text{sample } \mathcal{S}_{\text{base}}^h, \quad (5)$$

so that β^h is interpreted per 1%p of tariff reduction. Reading β^h this way adds a linear dose-response restriction, namely that the effect of an onset is proportional to

¹⁴The $j = 0$ component is already implied by $\Delta D_{it} = 1$.

G_i ; absent strict proportionality, β^h retains its interpretation as an intensity-weighted average of event-level effects expressed per 1%p. For each onset, G_i is the difference between the applied tariff rate and the Quota Tariff rate, the tariff reduction the designation delivers.¹⁵

The resulting intensities, with the applied tariff rate and the Quota Tariff rate from which each is computed, are reported for all eleven ever-treated products in table 2; they run from 1.5%p for kiwifruit to 69.9%p for onion.

Table 2: Quota Tariff Treatment Intensity

Product	Applied Tariff Rate (%)	Quota Tariff Rate (%)	Treatment Intensity (%p)
Napa cabbage	27.0	0	27.0
Cabbage	26.9	0	26.9
Radish	29.4	0	29.4
Carrot	28.7	0	28.7
Onion	79.9	10	69.9
Green onion	27.0	0	27.0
Mango	15.7	0	15.7
Kiwifruit	6.5	5	1.5
Avocado	8.0	0	8.0
Pineapple	29.6	0	29.6
Banana	24.5	0	24.5

The impulse specifications estimate the same equations (2) and (5). Only the sample changes: the stay-on condition is removed from the treated side, and the control side is unchanged:

$$\begin{aligned}
\mathcal{S}_{\text{imp}}^h = & \underbrace{\left\{ (i, t) : \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120] \right\}}_{\text{clean treated: fresh entry, subsequent path unrestricted}} \\
& \cup \underbrace{\left\{ (i, t) : D_{is} = 0 \forall s \leq t, D_{i,t+j} = 0 \forall j \in [1, h] \right\}}_{\text{clean control: as in (3)}}, \quad h \geq 0, \quad (6)
\end{aligned}$$

with coefficients denoted β_{imp}^h . For $h < 0$, the future no-treatment condition on the control side, $D_{i,t+j} = 0 \forall j \in [1, h]$, is dropped, as in the persistent-treatment specification; the treated-side condition remains unchanged, since it contains no forward-

¹⁵Products with multiple onsets receive the average across their onsets, a single time-invariant scalar per product. Averaging over the pre-onset year smooths short-term fluctuations and keeps G_i pre-determined: rates realized after the onset, which could respond to the treatment itself, never enter. Section 3.4 details the applied tariff rate.

looking element to begin with:

$$\mathcal{S}_{\text{imp}}^h = \underbrace{\left\{ (i, t) : \Delta D_{it} = 1, D_{i,t-s} = 0 \forall s \in [1, 120] \right\}}_{\text{clean treated: unchanged}} \cup \underbrace{\left\{ (i, t) : D_{is} = 0 \forall s \leq t \right\}}_{\text{clean control: no history through } t}, \quad h < 0. \quad (7)$$

This coincides exactly with the persistent-treatment sample in equation (4). At horizons $h < 0$, the two specifications therefore share the same sample and the same regression, which makes their pre-event coefficient paths identical; this coincidence extends to $h = 0$ as well.

The intensity scaling in equation (5) is the one element that goes beyond the binary framework of Dube et al. (2025), and its interpretation follows the dose framework of De Chaisemartin and D’Haultfœuille (2026).

Standard errors are clustered by product throughout, and 95% confidence bands use the normal critical value, matching the construction of the figures. Two caveats are in order. First, the number of *treated* clusters is small (six vegetable and five fruit products), and cluster-robust intervals at long horizons therefore deserve caution (MacKinnon and Webb, 2018); section 5.1 examines the issue and the available alternatives in detail. Second, at far horizons of the persistent-treatment specification the treated set itself thins (section 4.2). Both cautions inform the reading of the results.

The covariate vector X_{it} differs between the two treated groups, each estimated in a separate regression. The two comparison pools share a common base, the 30 never-treated products, supplemented in each case with the pre-onset observations of the group’s own ever-treated products.¹⁶ For the vegetable group,

$$X_{it}^{\text{veg}} = A_{it} + I_i \cdot O_t + I_i \cdot W1_{it} + I_i \cdot W2_{it} + R_{it} + Q_{it},$$

where A_{it} is the applied tariff rate (the tariff that would prevail absent the Quota Tariff; section 3.4.1), O_t is the oil price in KRW interacted with product dummies I_i to allow heterogeneous transport-cost channels, and $W1_{it}$ and $W2_{it}$ are climate

¹⁶Treated Group 1 comprises the six leafy and root vegetables, and Treated Group 2 consists of the five fruits listed in section 3.1; when one group is estimated, the other group’s products leave the sample. The two groups respond in opposite directions (section 4.5), so a pooled coefficient would average a negative fruit response against a positive vegetable response, concealing both.

covariates in the spirit of [Roberts and Schlenker \(2013\)](#): station averages of daily temperature, humidity, precipitation, and sunshine over $[t - 100, t]$ ($W1_{it}$), and the same averages one year earlier (over $[t - 465, t - 365]$, $W2_{it}$), each interacted with product dummies. In that account current climate shifts supply directly, while the previous year’s climate shapes planting decisions and hence current-year supply. R_{it} and Q_{it} collect the government stockpile-release and WTO market-access-quota covariates of section 3.6. For the fruit group the climate blocks are omitted:

$$X_{it}^{\text{fruit}} = A_{it} + I_i \cdot O_t + R_{it} + Q_{it}.$$

The treated fruits are predominantly imported, so Korean weather is not the relevant supply shifter for them, whereas domestic weather drives both vegetable supply and the timing of vegetable designations. Empirically, fruit pre-event coefficients remain flat without the climate covariates, whereas removing them from the vegetable regressions re-opens the pre-event run-up documented in section 4.4 below (figure 2).

Identification rests on the two assumptions standard for LP-DiD: no anticipation, and parallel trends of treated products relative to clean controls, conditional on X_{it} , the calendar-day effects, and product fixed effects.¹⁷ Anticipation has little room to operate, since most designations take effect with no advance legislative notice at all or with a lead of only one to five days, and a lead approaching one month occurred in just a few cases (section 3.3). The pre-event coefficients reported in section 4.5 speak directly to both assumptions. The LP-DiD results can be interpreted causally only if the conditional parallel-trends assumption holds, so the next subsection takes it to the data.

4.4 Pre-Designation Price Dynamics

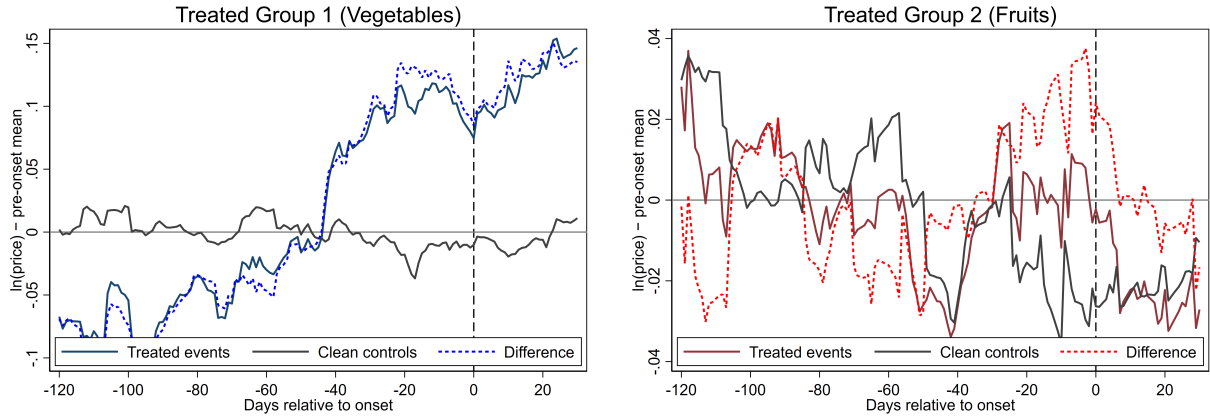
The Quota Tariff is, by statutory design, aimed at products whose prices have surged, which may place treated products on price paths unlike those of the comparison pool precisely when treatment begins. The concern is an Ashenfelter dip in reverse ([Ashenfelter, 1978](#)). If designations are triggered by rising prices, a later decline occurring for reasons unrelated to the tariff cut could be mistaken for an effect of the policy.

¹⁷Product fixed effects are absent from equations (2) and (5) by construction rather than omission: the dependent variable of equation (1) already subtracts each product’s own pre-event mean.

This subsection documents how retail prices move around designations and shows the vegetable run-up to be adequately absorbed by the climate covariates.

Figure 1 aligns every designation event on its onset date and traces average retail prices in event time, in the metric the estimation uses: the log retail price minus its own average over the 120 days preceding onset.¹⁸ The same transformation is applied to the 30 never-treated products on the same calendar dates, which lets market-wide movements enter both curves symmetrically. For vegetables the pattern is stark: prices climb from about 7 log points below the never-treated comparison four months before designation to 9~14 log points above it in the final three weeks, and the climb appears in eight of the nine events. For fruits there is no counterpart, the two curves staying within about 4 log points of each other throughout.

Figure 1: Retail Prices around Quota Tariff Onsets in Event Time

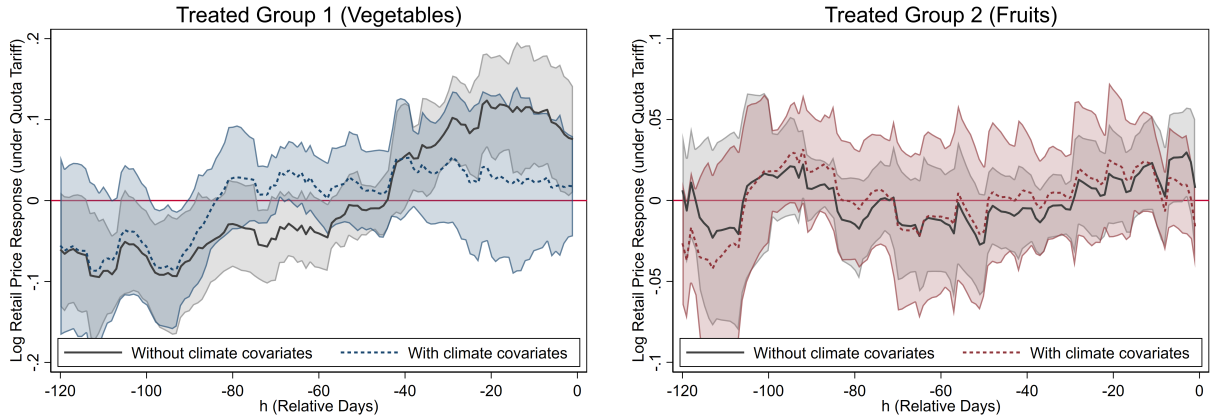


Whether this movement threatens identification depends on whether it survives the estimation's conditioning. Figure 2 reports event-study coefficients at pre-onset horizons $h \in [-120, -1]$ from the binary specification in equation (2), estimated with and without the climate blocks $I_i \cdot W1_{it}$ and $I_i \cdot W2_{it}$ on a common sample. Without the climate covariates the vegetable path reproduces that run-up in coefficient form: a significantly negative cluster deep in the pre-onset window, then significantly positive coefficients on the 39 consecutive days preceding onset, between +5 and +12

¹⁸The events are those that enter the estimation, namely onsets preceded by at least 120 treatment-free days (section 4.2): nine vegetable onsets and eight fruit onsets. By construction each series averages to zero over its own 120-day pre-onset window, here and in figure 2; the diagnostic content lies in the path within the window, not its level.

log points. With the climate blocks, adopted on supply-shifter grounds (Roberts and Schlenker, 2013), the same path collapses toward zero: 11 of 120 coefficients significant, none in the 66 days preceding onset, near-onset magnitudes of 2~5 log points. The coefficients themselves move toward zero rather than merely losing precision, and the gap between the two paths is statistically significant rather than sampling noise. Because both are estimated on the same sample, the paths can be differenced horizon by horizon on a grid fixed in advance ($h = -120, -105, \dots, -30$), with the standard error of the difference from a cluster jackknife that leaves out one product at a time; the test rejects the hypothesis that they coincide ($t = -2.0$ for the average difference across the grid, $t = -2.9$ at $h = -75$). For the fruit group the two versions are indistinguishable ($t = 0.00$) and both are flat, with 1~2 of 120 coefficients significant, which confirms leaving the climate blocks out of the fruit specification. The parallel-trends assumption of section 4.3 is *conditional on the covariates*, and the graphical results in this subsection show that the systematic pre-designation differences between the vegetable group and the comparison group are absorbed by those covariates. Subject to the two caveats of section 4.3, the analysis turns to the estimates.

Figure 2: Pre-Onset Coefficients with and without Climate Covariates



4.5 Graphical Results

Figures 3, 4, 5, and 6 present the four sets of estimates: the treatment-status specification in its persistent-treatment and impulse-response forms, then the treatment-intensity specification in the same order. In all four, solid blue lines denote Treated

Group 1 (vegetables), dashed red lines denote Treated Group 2 (fruits), and the shading indicates 95% confidence intervals (clustered by product). The treatment-intensity estimates are the main results of this study: a Quota Tariff onset lowers the tariff rate by 1.5%p to 69.9%p depending on the product, and the intensity specification accounts for that depth, whereas the treatment-status specification captures the adoption status regardless of the depth of the cut.

Figure 3: Treatment Status (Binary), Persistent-Treatment Specification

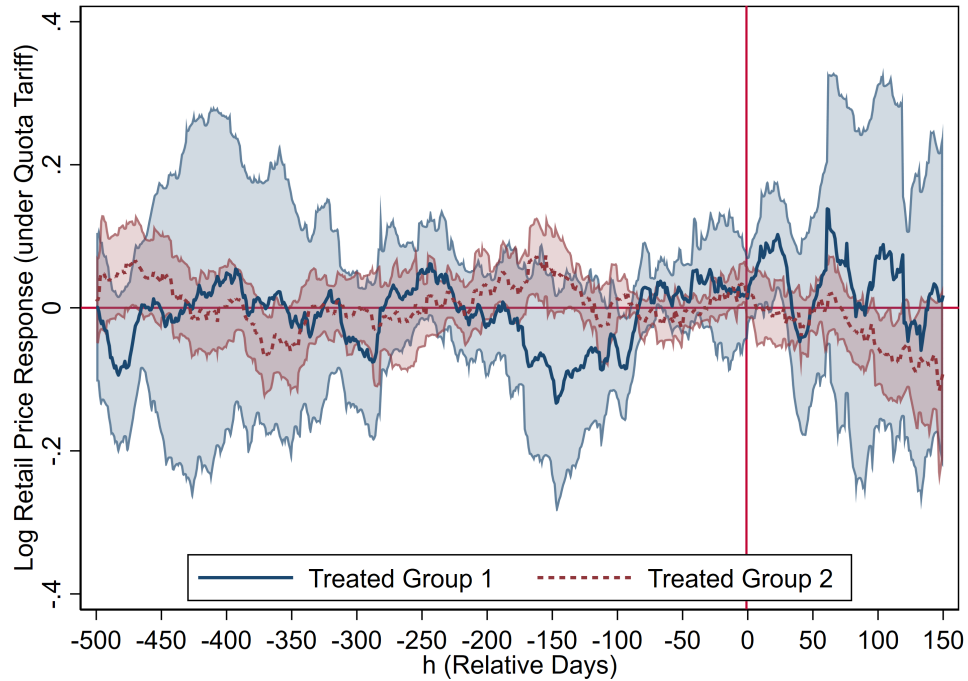


Figure 3 shows the treatment-status estimates under persistent treatment. The pre-event window supports the identifying assumptions in both groups. For vegetables (Treated Group 1), 11 of the 120 pre-event coefficients are significant at the 5% level, modestly above the six expected by chance, but they sit in a single nine-day negative cluster at $h \in [-99, -91]$ and two isolated positive days ($h = -69$ and $h = -67$), none within the 66 days preceding adoption. For fruits (Treated Group 2), only 5 of 120 are significant, below the six expected by chance, and all five form a short positive run at $h \in [-7, -3]$ of roughly +3 to +4 log points, too small and brief to disturb a reference level averaged over 120 pre-event days (equation (1)).

Post-onset the two groups diverge. Vegetable prices do not fall: the coefficients are never significantly negative, turn significantly positive in the first weeks after adoption (at $h \in [8, 10]$, $h = 14$, and $h \in [20, 27]$, reaching about +10 log points at $h = 23$), and are insignificant thereafter through $h = 150$. Those early positives should not be read as the tariff cut raising prices: if the cut were the cause they would be larger where cuts are deeper, yet they vanish in the intensity specification below.

Fruit prices, in contrast, begin to fall about three months after adoption: significant negative pockets open over $h \in [90, 131]$ (about -5.7 log points, or -5.5 percent, at $h = 100$) and recur through $h \approx 200$ in the replication files.¹⁹ Treating shallow-cut events such as kiwifruit (1.5%p) on the same footing as deep-cut events such as banana (about 25%p) makes the fruit response hard to estimate precisely; the treatment-intensity specification reveals it far more clearly.

Figure 4: Treatment Status (Binary), Impulse-Response Specification

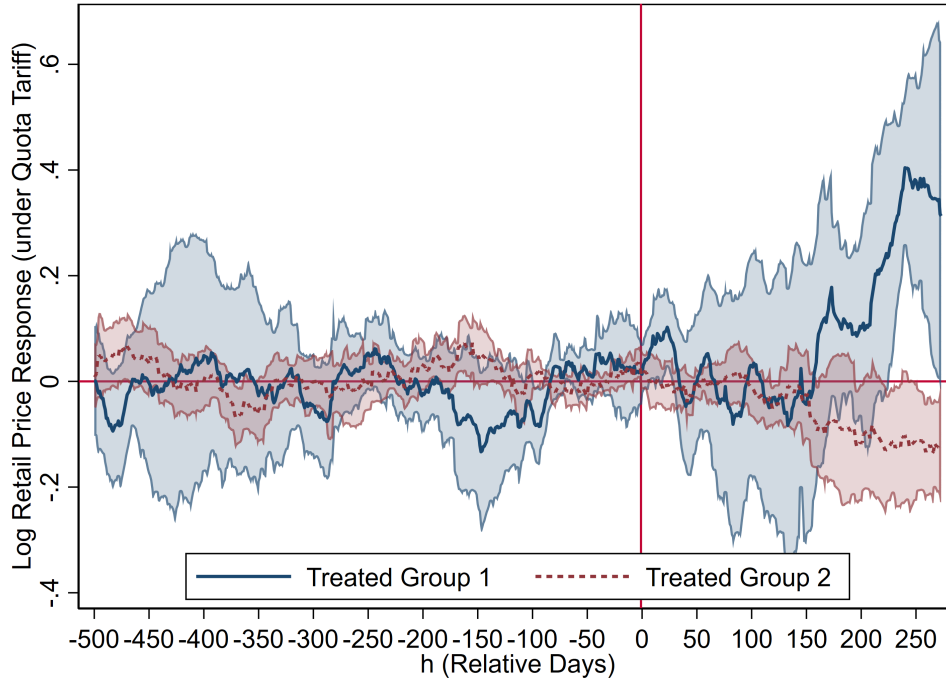


Figure 4 gives the impulse-response form of the treatment-status specification, the total effect of initiation, tracked to $h = 273$ with no attrition of treated products. For

¹⁹However, the estimate at $h = 150$ (-8.9 log points) is imprecise.

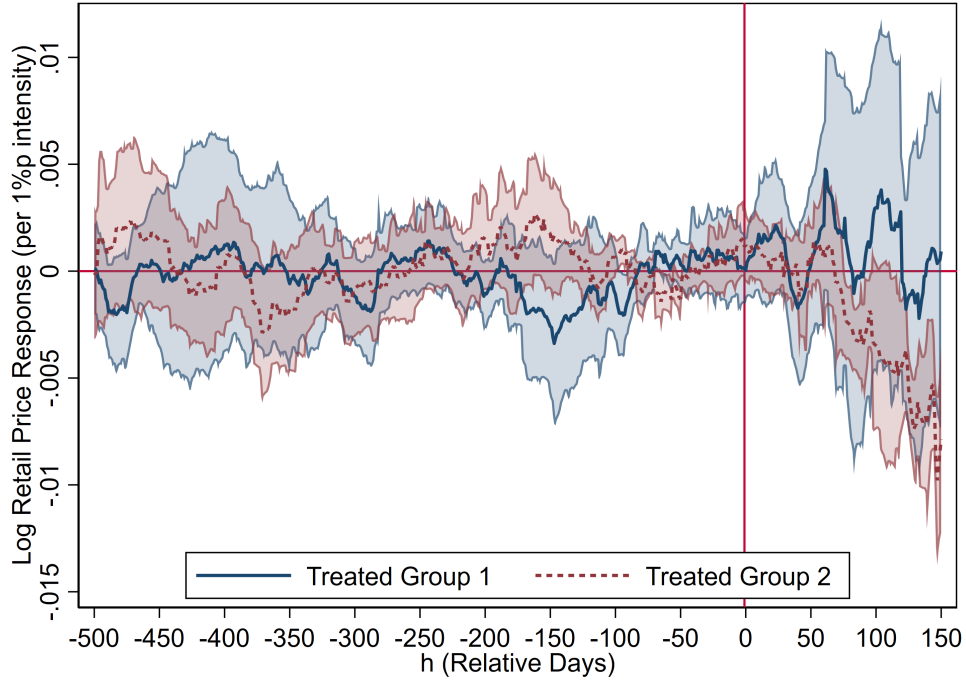
fruits (Treated Group 2), the initiation effect turns significantly negative in pockets from $h = 118$ onward (about -4.5 log points over $h \in [118, 124]$) and settles into a sustained decline at far horizons: -11.6 log points at $h = 200$ and -11.5 at $h = 250$, remaining there through $h = 273$, a decline of roughly 11 percent. Since many fruit episodes have ended well before these horizons, the persistence of the decline indicates that the effect of Quota Tariff initiation outlives the period during which the treatment remains in force. For vegetables, apart from the early positive coefficients already seen under persistent treatment,²⁰ the estimates are insignificant through $h \approx 225$ and then turn sharply positive ($+38$ log points, or $+46\%$, at $h = 250$). This far-horizon vegetable estimate should not be interpreted causally (the impulse-response specification imposes no stay-on condition, so the treated set is complete at every horizon here, which rules out a composition artifact). It is driven instead by the four events initiated in 2024: napa cabbage, cabbage and carrot in May, radish in July. Their 200~270-day windows fall between November 2024 and March 2025, when the kimchi-vegetable markets, radish and napa cabbage above all, stood far above their pre-designation levels in the wake of the autumn-2024 supply disruptions. These far-horizon vegetable coefficients pick up a supply disturbance specific to these products and are reported for transparency but excluded from the causal conclusions.

Figure 5 turns to the treatment-intensity specification, the persistent-treatment response per 1%p of the tariff reduction a Quota Tariff onset induces. For vegetables (Treated Group 1) the pre-event record is clean (5 of 120 coefficients significant, none within 67 days of onset), and no significant response appears anywhere through $h = 150$. Under ongoing treatment, vegetable prices simply do not fall per unit of tariff reduction. For fruits (Treated Group 2), 17 of 120 pre-event coefficients are significant, above the chance rate, and this is the one pre-event diagnostic in the four figures that warrants scrutiny. The significant days fall in two clusters (positive around $h \in [-99, -92]$, negative around $h \in [-65, -50]$), both seven weeks or more before adoption, with none in the 49 days preceding the onset. Since adjacent horizons are serially correlated, clusters like these can arise; adjacent to the Quota Tariff onset there is no systematic pattern of significant coefficients.

Post-onset the fruit response hovers near zero for the first ten weeks and is negative essentially throughout thereafter. It is significant at $h \in [76, 96]$ (roughly -0.2 to

²⁰The two samples are mechanically identical at short horizons, before any spell has ended.

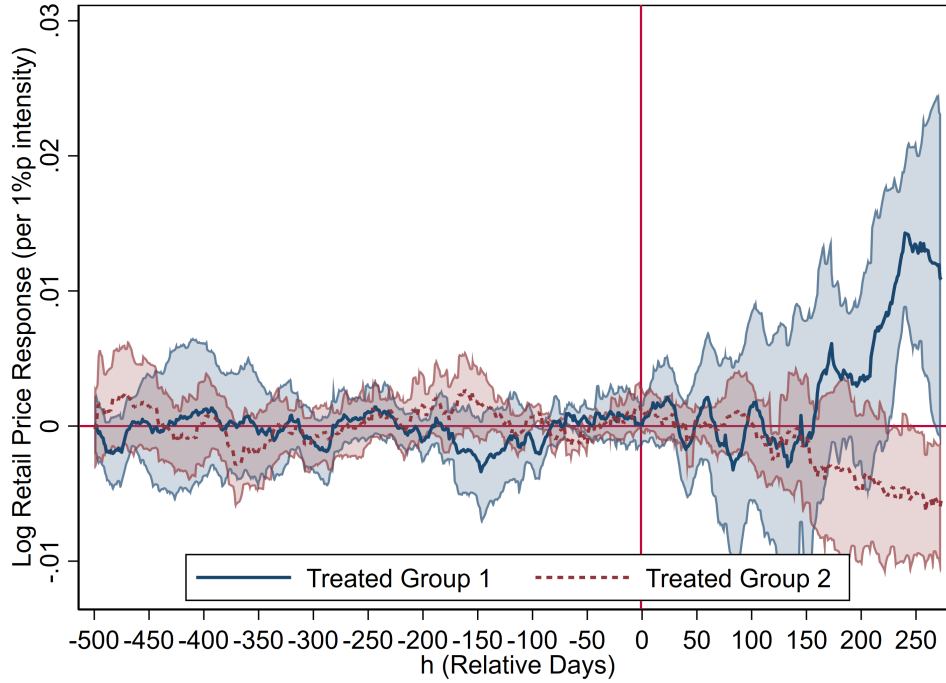
Figure 5: Treatment Intensity (per 1%p), Persistent-Treatment Specification



-0.35 per 1%p, in log points, i.e. the coefficient $\times 100$) and reaches -0.79 at the plotted window's edge, $h = 150$. Beyond that window the fruit treated set is unchanged through $h = 270$ (no further event exits until $h = 271$), and the replication files show the significant band continuing over $h \in [112, 209]$ and reaching -1.5 per 1%p at $h = 182$. The graphs stop at $h = 150$ only because of the attrition of treated events in the vegetable group. This significant fall in fruit retail prices between 90 and 210 days after onset, at economically meaningful magnitudes per 1%p of tariff reduction, is the core causal finding of the paper, and the sign of the response is robust across all four specifications; section 5.1 returns to the mid-horizon magnitudes, which exceed what full pass-through would imply.

Figure 6 completes the set. For fruits (Treated Group 2), the impulse responses of the treatment-intensity specification repeat the pattern of their treatment-status counterparts: significant negative pockets from $h = 119$ onward (-0.23% per 1%p over $h \in [119, 124]$), then a sustained significant decline at far horizons, -0.49% per 1%p at $h = 250$ and -0.55% at $h = 273$. For vegetables (Treated Group 1) the

Figure 6: Treatment Intensity (per 1%p), Impulse-Response Specification



estimates are insignificant until $h = 226$ and then jump abruptly positive (+1.35% per 1%p at $h = 250$), subject to the caution noted above for the treatment-status impulse responses.

Two diagnostics complete the review. The first is the reverse Ashenfelter dip of section 4.4, against which three features speak: the pre-event coefficients are flat in the window adjacent to adoption in all four specifications;²¹ equation (1) anchors the reference level to a 120-day average that a surge in the last few days barely moves, while the calendar-day fixed effects absorb market-wide surges; and the vegetable climate covariates absorb the supply shocks behind both price surges and designations (figure 2). A classic negative dip is likewise absent. The only case that cannot be fully excluded is anticipation happening to offset a dip exactly, but such a coincidental offset is highly unlikely because, as described earlier, anticipation has little room to operate.

The second diagnostic is economic magnitude. Under full and immediate pass-through, a 1%p tariff reduction lowers the consumer price by approximately $100/(100+$

²¹The sole exception is the five-day blip for fruits discussed above.

$\bar{\tau}$) percent, where $\bar{\tau}$ is the prevailing tariff level; for the treated fruit products the counterfactual applied tariff rate $\bar{\tau}$ ranges from about 6.5 to 29.6 percent (table 2), placing the full-pass-through benchmark between roughly 0.77 and 0.94 percent per 1%p. The estimated fruit responses (-0.78% per 1%p at $h = 150$ under ongoing treatment, and -0.49% to -0.55% per 1%p at $h \geq 250$ for initiations) therefore imply substantial, at some horizons near-complete, pass-through of the tariff reduction to consumers.²² The magnitude varies with horizon and specification, and these estimates bracket the plausible range rather than pin down a single pass-through rate.

4.6 LP-DiD Estimates at Selected Horizons

Table 3 condenses the four sets of estimates into regression form at horizons chosen by the design, not by the results. For the persistent-treatment specification, $h = 150$ is the truncation point of its graphs, a conservative margin below the compositional break at $h = 156$ and the longest horizon at which its results are reported. The impulse-response specification is reported at the same horizon for comparability and at $h = 250$, roughly eight months after initiation, where its treated sets are still complete (nine vegetable and eight fruit events). Coefficients and standard errors are multiplied by 100 for readability.

The table restates the four figures in compact form. At $h = 150$ in the persistent-treatment specification, the only significant estimate is the fruit response per 1%p of intensity, -0.79 (in log points, i.e. the coefficient $\times 100$; panel A, column 4); the fruit estimate in the treatment-status specification is negative but insignificant (panel A, column 2), and both vegetable estimates (panel A, columns 1 and 3) are near zero. At the same horizon the impulse-response estimates (panel B) are insignificant throughout. Contrasting panel B with panel C shows how long the effect of initiation takes to materialize: by $h = 250$ it emerges as significant fruit declines (-11.5 binary, panel C, column 2; -0.50 per 1%p, panel C, column 4). As detailed above, the far-horizon vegetable positives in that panel (panel C, columns 1 and 3) reflect the contemporaneous supply disturbance in the products designated in 2024 rather than causal effects;

²²Computed as $(e^{\beta^h} - 1) \times 100$: -0.783% at $h = 150$ (persistent treatment), -0.494% at $h = 250$ and -0.548% at $h = 273$ (impulse). The benchmark treats the quota as non-binding within the window and abstracts from inventory and contracting lags, making it an upper bound on the speed, not the existence, of pass-through.

Table 3: Local Projection Difference-in-Differences Estimates at Selected Horizons

	(1) Quota Tariff Treatment Status	(2) Quota Tariff Treatment Status	(3) Quota Tariff Treatment Intensity (1%p)	(4) Quota Tariff Treatment Intensity (1%p)
	Group 1 (Vegetables)	Group 2 (Fruits)	Group 1 (Vegetables)	Group 2 (Fruits)
<i>Panel A: Persistent treatment, $h = 150$</i>				
Quota Tariff Shock	1.740 (12.260)	-8.950 (6.166)	0.092 (0.420)	-0.786*** (0.224)
Observations	33678	45026	33678	45026
R^2	0.492	0.159	0.492	0.159
Adjusted R^2	0.471	0.131	0.471	0.131
<i>Panel B: Initiation (impulse), $h = 150$</i>				
Quota Tariff Shock	-4.360 (13.521)	-2.338 (4.400)	-0.143 (0.476)	-0.088 (0.247)
Observations	33682	45030	33682	45030
R^2	0.492	0.159	0.492	0.159
Adjusted R^2	0.470	0.131	0.470	0.131
<i>Panel C: Initiation (impulse), $h = 250$</i>				
Quota Tariff Shock	37.966*** (10.634)	-11.519** (5.653)	1.340*** (0.397)	-0.495** (0.237)
Observations	30154	41530	30154	41530
R^2	0.593	0.191	0.593	0.191
Adjusted R^2	0.576	0.164	0.576	0.164

Coefficients and standard errors multiplied by 100 for readability; standard errors, in parentheses, clustered by product

(36 clusters in columns 1 and 3, 35 clusters in columns 2 and 4). Treated clusters differ by panel because

the stay-on condition removes events as h grows: panel A rests on 5 treated vegetable and 4 treated fruit products, panels B and C on 6 and 5, the full ever-treated sets.

Significance is based on the normal critical values used for the confidence bands in the figures.

Panel A conditions on treatment persisting through h ; Panels B and C track initiations regardless of the subsequent path.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

the impulse-response treated set is complete at $h = 250$, leaving no composition effect involved. The explanatory power differs mechanically across groups.²³ The modest R^2 values themselves have no bearing on causal identification.²⁴

Taken together, the causal analysis yields one robust policy-relevant effect and one clearly labeled non-effect. Quota Tariff designations on imported fruits lower retail prices per 1%p of tariff reduction, significantly and persistently, with the decline outliving the designations themselves; designations on domestic leafy and root vegetables, by contrast, show no detectable price-reducing effect at any interpretable horizon, under either estimand. The next section probes the robustness of these conclusions.

²³The vegetable regressions, with their product-specific climate blocks, absorb roughly half of the variance; the leaner fruit specification absorbs less.

²⁴The modest R^2 values presumably arise because each dependent variable is a long-horizon daily deviation from a 120-day pre-event mean.

5 Robustness

5.1 Inference with Few Treated Clusters

The number of treated clusters is set by the policy itself: over the sample period the Quota Tariff reached six vegetable and five fruit products, and the stay-on condition leaves five and four of them in the persistent-treatment regressions at $h = 150$. Any estimator here must accordingly conduct inference with few treated clusters, where cluster-robust t -statistics tend to over-reject because they contribute too few residual blocks for their variance to be estimated reliably (Conley and Taber, 2011; MacKinnon and Webb, 2020). Corrections push the other way and overshoot: the restricted wild cluster bootstrap under-rejects in exactly this configuration, its most recent variants severely so when the number of treated clusters is small (MacKinnon et al., 2023), and its wide intervals therefore reflect the documented failure of the procedure rather than sampling uncertainty in the estimates. MacKinnon and Webb (2018) conclude that when the restricted and unrestricted bootstrap p -values diverge, as they systematically do with few treated clusters, neither can be trusted.

Alternatives exist, but each restores validity by imposing a homogeneity assumption that the present setting violates by construction; Roth et al. (2023) summarize that “all of the available approaches require the researcher to impose some potentially strong additional assumptions.” Appendix G reviews the randomization, placebo, re-scaling and wild-bootstrap alternatives one by one and shows why none applies here; all are, moreover, developed for a binary treatment that turns on once and stays on, which places a non-absorbing, continuous-intensity design of the kind the Quota Tariff requires outside their scope.

The analysis therefore reports conventional product-clustered intervals and disciplines their reading three ways. First, the documented distortion of cluster-robust t -statistics with few treated clusters is over-rejection, a bias toward finding effects; the conclusion for vegetables is thus reached against that bias and is all the more robust: the null of no effect was not rejected even by tests strongly predisposed to reject it. The far-post vegetable positives, the one place where a liberal test could mislead, are already excluded on confounding grounds (section 4.5). Second, the fruit effect does not rest on an isolated significant coefficient of the kind small-sample noise produces:

it appears as contiguous bands of horizons, with the same sign and timing in all four specifications. At the horizons reported in table 3 its magnitude sits at or below what a tariff pass-through rate of 100% would imply, while the larger mid-horizon estimates reported above exceed that benchmark and therefore cannot be attributed to the tariff reduction alone. Third, following the reporting guidance of [MacKinnon et al. \(2023\)](#), the number of clusters and the treated-cluster counts are reported with the estimates in table 3, and the event-level composition of the treated sets, together with the horizons at which events exit them, is disclosed in Appendix F.

6 Conclusion and Policy Implications

This paper identifies the causal effect of Korea’s Quota Tariff on agricultural retail prices. Because a designation lapses when its announced period ends or its quota is exhausted, and the same product can be designated again, the treatment is non-absorbing. Under that structure conventional TWFE event studies draw comparisons from already-treated observations, while the leading heterogeneity-robust estimators for staggered adoption, which presuppose absorbing treatment, do not apply. Estimating it on the switching path itself, with LP-DiD under clean controls and a 120-day washout, is the methodological contribution, and the treatment-intensity specification extends that design to a treatment that is simultaneously non-absorbing and continuous.

Two findings follow. First, the response is sharply heterogeneous: for leafy and root vegetables no price-reducing effect is detectable at any interpretable horizon under either estimand, whereas for fruits prices fall by about 0.78% per 1%p of tariff reduction at $h = 150$ while the reduction is in force, and by 0.49% to 0.55% per 1%p at $h \geq 250$ for initiations. Set against the 0.77~0.94% per 1%p implied by full and immediate pass-through, the fruit estimates indicate that a substantial share of the reduction reached consumers, at the shorter horizon almost all of it. Second, the effect for fruits emerges only after several months and persists past the designations themselves—a separation the persistent-treatment and impulse-response estimands are built to reveal. Where the policy works, it works on a medium-term horizon rather than on impact.

The heterogeneity tracks supply structure. The treated fruits are predominantly imported, which gives a tariff reduction a direct channel into retail prices, whereas that

channel is structurally limited for most of the treated vegetables, produced largely at home. The policy implication is therefore selective rather than general: since the Quota Tariff carries a direct fiscal cost in forgone tariff revenue, concentrating designations on products with a high import share and reconsidering those on vegetables would deliver a larger price-stabilization effect at the same cost.

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A Appendix: Summary Statistics

This appendix summarizes the estimation sample and variables in three tables. Table 4 reports descriptive statistics for the variables entering the LP-DiD regressions, while table 5 and table 6 report the annual import scale of the 41 products. Table 4 is organized in three panels. Panel A covers the daily panel variables: the seasonally adjusted retail price from which the dependent variable is built, the daily Quota Tariff treatment status, and the covariates for the applied tariff rate, climate, oil prices, and WTO Market Access Quotas. For the two variables in panel B, summary statistics are computed not at the daily level but at the unit at which each variable enters the LP-DiD regressions. Treatment intensity G , the tariff reduction induced by a designation, varies across ever-treated products but not over time,²⁵ and its summary statistics are therefore computed at the product level. Government stock releases are recorded as monthly totals in the raw data, which places their summary statistics at the product-month level. Panel C reports the seasonally adjusted retail price of each of the 41 products in four blocks: treated leafy and root vegetables, treated fruits, never-treated vegetables and other crops, and never-treated fruits. The reported statistics (N, mean, standard deviation, minimum, and maximum) are the standard set in the empirical pass-through literature.

All entries in table 4 are level values in their raw units, prior to any transformation for use in the LP-DiD regressions. In the estimation itself, the dependent variable is the log of the seasonally adjusted retail price differenced against its average over the 120 days preceding the event, as defined in equation (1), and the covariates enter in the forms defined in sections 3.6 and 4.3. Reporting raw levels keeps the scale and range of each variable visible rather than hidden behind those transformations.

Table 4: Summary Statistics of Regression Variables

	N	Mean	SD	Min	Max
<i>Panel A. Daily panel variables</i>					
Retail price, seasonally adjusted (KRW/kg)	63,013	9,435.9	7,494.7	753.5	36,797.0
Quota Tariff treatment status (0/1)	63,591	0.062	0.240	0.000	1.000

continued on next page

²⁵Treatment intensity G —the difference between the tariff rate that would apply in the absence of the designation (the applied tariff rate) and the rate under the designation—is indeed time-varying, since the applied tariff rate changes monthly. However, by construction the LP-DiD model requires a time-invariant treatment variable, and because G in this study is defined as the tariff reduction at the time the designation takes effect, we hold G fixed at that value.

Table 4 (continued)

	N	Mean	SD	Min	Max
Applied tariff rate (%)	63,591	35.5	42.5	0.0	587.9
Mean temperature (°C)	63,550	13.3	10.8	-14.9	31.8
Mean relative humidity (%)	63,591	65.2	13.8	22.9	99.3
Precipitation index	63,345	2.2	0.9	0.1	3.8
Sunshine duration (hours)	63,345	6.2	4.0	0.0	13.5
Real crude oil price (×1,000)	63,591	20.7	3.9	14.2	34.9
WTO market access quota, applicable (0/1)	63,591	0.341	0.474	0.000	1.000
WTO quota expansion (ratio)	63,591	0.276	1.420	0.000	9.548
WTO expansion intensity	63,591	1.441	8.258	0.000	56.334
<i>Panel B. Treatment intensity and policy variables</i>					
Treatment intensity G (%p; ever-treated items)	11	26.2	17.3	1.5	69.9
Government stock releases (tons/month; item×month)	2,091	70.0	388.9	0.0	5,214.0
<i>Panel C. Seasonally adjusted retail price (KRW/kg), by product</i>					
Treated: leafy and root vegetables					
cabbage	1,548	2,180.8	331.4	1,504.5	3,061.4
carrot	1,548	4,060.7	820.0	2,703.9	6,072.3
green onion	1,548	3,130.0	825.9	1,711.7	6,361.5
napa cabbage	1,548	2,253.6	404.8	1,496.8	3,758.7
onion	1,548	2,146.6	363.4	1,505.0	3,088.1
radish	1,548	1,222.6	250.4	753.5	1,974.4
Treated: fruits					
avocado	1,093	8,359.5	988.9	6,385.9	11,981.9
banana	1,548	2,851.3	300.3	1,956.4	3,346.3
kiwifruit	1,548	4,714.2	323.4	3,877.7	5,558.7
mango	1,548	13,505.1	2,057.3	8,666.4	19,104.1
pineapple	1,548	4,625.3	342.0	3,663.3	5,500.4
Never-treated: vegetables and other crops					
adzuki bean	1,548	16,149.5	2,725.4	12,131.5	24,999.6
bean	1,548	9,937.2	1,107.9	7,743.0	12,224.3
bell pepper	1,548	8,176.9	781.8	6,014.7	11,121.2
cherry tomato	1,548	8,345.8	836.9	6,072.9	10,715.3
cucumber	1,548	4,071.2	555.6	2,817.7	5,941.3
dried red pepper	1,548	26,393.4	3,793.7	20,620.9	36,797.0
garlic	1,548	10,297.2	1,983.8	6,741.8	13,866.9
ginger	1,548	11,304.9	2,363.3	7,155.3	16,390.4
green pepper	1,548	11,641.5	1,826.2	7,999.8	18,859.5
lettuce	1,548	11,134.9	1,917.4	7,419.2	18,435.2
mung bean	1,548	20,505.6	1,211.9	18,071.3	23,180.6
peanut	1,548	27,732.3	2,102.0	24,094.2	33,784.4
perilla leaf	1,548	23,060.0	2,337.6	17,495.5	29,706.7
red pepper	1,548	18,386.9	2,860.9	11,551.6	29,229.2
scallion	1,548	7,672.0	1,171.0	5,059.3	10,891.7
sesame	1,548	31,001.2	2,340.7	26,306.4	36,493.6
soybean sprout	1,548	4,572.9	487.7	3,136.4	5,585.0

continued on next page

Table 4 (continued)

	N	Mean	SD	Min	Max
spinach	1,548	9,404.9	1,544.8	5,566.8	15,590.7
squash	1,548	5,410.5	892.2	3,612.6	9,152.9
sweet pepper	1,548	12,274.1	1,731.0	8,625.9	18,976.1
sweet potato	1,548	5,159.0	716.9	3,927.9	6,941.1
tomato	1,548	5,516.2	682.9	3,919.8	7,626.1
water dropwort	1,548	11,250.7	3,194.3	6,435.8	17,156.8
young napa cabbage	1,548	2,941.1	405.6	2,063.8	4,214.5
young summer radish	1,548	3,220.3	726.4	1,956.0	6,398.6
Never-treated: fruits					
apple	1,548	8,779.5	1,114.7	6,599.0	12,383.1
lemon	1,548	6,747.0	457.0	5,584.8	7,976.3
melon	1,548	6,352.3	584.9	5,112.6	7,785.4
pear	1,548	5,875.6	1,180.7	3,200.4	9,583.2
watermelon	1,548	4,192.1	351.5	3,211.7	4,899.9

The estimation panel covers 41 products from January 1, 2021 to March 31, 2025. The retail-price series is seasonally adjusted (Appendix E); the panel is unbalanced because avocado’s series begins on April 4, 2022, lowering its N . Treatment status is the raw daily Quota Tariff indicator, whereas the treatment path used in estimation bridges administrative gaps of at most 30 days (Appendix F). Treatment intensity G is a single time-invariant value per ever-treated product ($N = 11$), whose per-product values appear in Table 2; government stockpile releases are summarized at the product-month level. The real crude oil price is shown multiplied by 1,000 for readability; the underlying value equals real WTI divided by the KRW/USD exchange rate. In the regressions the climate variables enter as $[t - 100, t]$ averages and their one-year lags, and stockpile releases enter through an inverse-hyperbolic-sine transform with an occurrence dummy (Section 3.6); the estimation sample varies by horizon under the clean-treated and clean-control conditions, and horizon-specific treated and control counts accompany the estimates.

Table 5 reports the annual import volume of the 41 products in metric tons, obtained by aggregating the monthly import records in the customs data to product-year totals. The rows follow the same four blocks and ordering as panel C of table 4. The salient contrast in the two import tables lies between the treated groups. The five treated fruits show substantial imports in every year of the sample period (banana, for example, at roughly 400,000 tons in 2024), whereas the treated vegetables, apart from carrot and onion, import on a far smaller scale and much more erratically. The never-treated products range from items with no recorded imports at all to large importers such as bean. This contrast underpins the interpretation in the main text that the treated fruits are predominantly imported, giving a tariff reduction a direct channel into retail prices, whereas the mostly domestically produced vegetables offer little structural scope for that channel.

Table 5: Annual Import Volume by Product

	2021	2022	2023	2024	2025 (Jan.–Mar.)
Treated: leafy and root vegetables					
cabbage	4,450.1	4,333.7	6,741.6	23,527.3	19,989.6
carrot	91,719.9	96,243.6	110,396.1	117,695.2	29,917.4
green onion	8,272.3	1,135.5	7,748.9	23,388.6	3,924.9
napa cabbage	66.7	2,030.1	164.2	4,135.2	5,383.4
onion	38,513.4	70,650.9	117,075.1	78,625.2	42,678.2
radish	7,693.8	2,461.1	1,307.3	14,973.3	8,700.2
Treated: fruits					
avocado	17,313.1	13,479.2	12,671.2	11,251.6	2,450.1
banana	351,902.9	319,852.0	329,025.1	400,769.4	95,363.4
kiwifruit	41,000.8	44,350.8	37,191.8	49,649.9	2,280.4
mango	21,618.4	22,471.4	24,829.7	32,094.9	11,312.3
pineapple	61,688.0	67,104.1	67,047.7	81,573.9	16,720.9
Never-treated: vegetables and other crops					
adzuki bean	18,523.7	20,587.5	21,653.7	23,153.4	5,288.6
bean	236,141.2	277,082.0	242,931.3	251,862.6	78,334.5
bell pepper	0.0	4.4	0.0	0	5.8
cherry tomato	0.0	0	0	0	0
cucumber	0	0	0	0.0	0
dried red pepper	3,809.0	3,937.0	6,458.9	5,184.7	1,012.0
garlic	1,022.1	1,428.9	330.2	425.0	122.0
ginger	5,640.7	2,799.4	4,081.2	4,462.9	1,358.0
green pepper	2.0	0.0	11.1	5.5	6.4
lettuce	6,676.2	3,365.9	5,043.2	11,080.7	2,849.1
mung bean	16,213.3	12,193.6	14,001.4	14,067.6	3,424.4
peanut	30,949.7	31,352.6	30,695.1	33,303.6	10,276.6
perilla leaf	10,458.1	0	0.0	0	0
red pepper	2.0	0.0	11.1	5.5	6.4
scallion	2,320.8	22.7	38.2	116.3	34.7
sesame	39,948.8	37,841.6	49,419.1	53,651.7	15,019.1
soybean sprout	10,458.1	9,125.0	9,515.5	11,170.9	3,376.8
spinach	0	0.0	0.1	0.1	0
squash	25,203.6	19,808.5	12,067.6	22,110.4	7,906.1
sweet pepper	0.0	4.4	0.0	0	5.8
sweet potato	0	0	0	0.0	0
tomato	0.0	0	0	0	0
water dropwort	10,458.1	9,125.0	9,515.5	11,170.9	3,376.8
young napa cabbage	66.7	2,030.1	164.2	4,135.2	5,383.4
young summer radish	22,918.0	22,383.3	23,229.9	21,597.4	4,603.6
Never-treated: fruits					
apple	0.0	0	0	0.0	0
lemon	17,348.3	20,466.7	19,710.4	28,037.2	6,624.2
melon	1,944.6	1,727.1	3,150.3	3,434.6	0.0
pear	18.5	12.8	0.0	10.4	13.6
watermelon	0	0	0	0	0

Volume in metric tons. “0” denotes no recorded imports, whereas “0.0” denotes positive imports below

50 kg. The 2025 column covers January–March only and is not comparable to full calendar years.

Under the same blocks and notation, table 6 reports the corresponding annual import values in thousands of USD, complementing the volume figures of table 5. Because unit values differ widely across products, the two tables do not rank products alike: kiwifruit, for example, imports less by weight than pineapple yet records substantially larger import values in each year of 2021~2024.

Table 6: Annual Import Value by Product

	2021	2022	2023	2024	2025 (Jan.–Mar.)
Treated: leafy and root vegetables					
cabbage	1,607	1,485	1,953	8,542	6,083
carrot	42,528	44,096	38,877	51,065	12,831
green onion	8,952	1,120	4,496	15,163	2,216
napa cabbage	45	1,140	77	2,290	1,824
onion	17,111	32,760	46,869	25,788	12,827
radish	1,521	673	435	3,227	3,079
Treated: fruits					
avocado	50,386	42,240	32,267	36,885	9,466
banana	290,149	283,876	304,728	382,031	77,219
kiwifruit	145,453	158,945	131,100	175,129	6,615
mango	90,655	97,045	98,373	138,524	42,950
pineapple	52,639	62,218	65,872	85,603	14,941
Never-treated: vegetables and other crops					
adzuki bean	37,175	43,531	34,537	32,895	7,759
bean	168,983	234,705	219,768	207,219	57,934
bell pepper	0	11	0	0	27
cherry tomato	0	0	0	0	0
cucumber	0	0	0	0	0
dried red pepper	10,654	11,842	23,566	16,418	2,876
garlic	1,836	2,342	636	934	257
ginger	7,915	1,378	6,900	5,628	1,592
green pepper	11	0	11	5	6
lettuce	5,337	2,967	4,969	9,738	2,113
mung bean	32,555	25,106	30,955	31,324	7,567
peanut	60,473	61,984	64,073	65,134	18,543
perilla leaf	18,476	0	0	0	0
red pepper	11	0	11	5	6
scallion	2,953	8	14	47	14
sesame	82,433	86,444	116,143	122,388	32,713
soybean sprout	18,476	18,562	18,456	16,898	3,715
spinach	0	0	1	1	0
squash	17,120	14,606	13,696	12,318	5,279
sweet pepper	0	11	0	0	27
sweet potato	0	0	0	0	0

continued on next page

Table 6 (continued)

	2021	2022	2023	2024	2025 (Jan.–Mar.)
tomato	0	0	0	0	0
water dropwort	18,476	18,562	18,456	16,898	3,715
young napa cabbage	45	1,140	77	2,290	1,824
young summer radish	12,474	12,729	12,144	16,834	4,146
Never-treated: fruits					
apple	1	0	0	0	0
lemon	38,678	43,260	42,271	58,915	13,167
melon	2,920	3,149	5,152	5,752	0
pear	90	71	0	49	56
watermelon	0	0	0	0	0

Value in thousands of USD, rounded to the nearest thousand; “0” denotes no recorded imports or a positive value below USD 500. The 2025 column covers January–March only and is not comparable to full calendar years.

B Appendix: Quota Exhaustion Dates for Pineapple

This appendix works through the exhaustion calculation for each of the six pineapple designations. In table 7, Start date and End date delimit the designated period announced in advance by the government, irrespective of any subsequent quota exhaustion, and Quota (ton) is the quota announced alongside it. For No. 1, Method A shows cumulative imports falling short of the quota through the End date, whereas Method B places the crossing on December 21, 2022. Since Method B overstates volumes, the authoritative Method A record prevails, and pineapple is taken never to have exhausted its quota during the No. 1 period. For No. 2, Method A establishes that the quota was exceeded at some point in October 2023, because no imports cleared under the Quota Tariff are recorded for November or December; Method B dates the crossing to October 9, 2023, and this date is adopted. Reflecting this, the cells from October 9, 2023 onward within the No. 2 period are white in table 1 (not-treated status).

For No. 3, Method A records imports cleared under the Quota Tariff all the way to the End date. Method B, computed against the initially announced quota of 40,000 tons, places the crossing of cumulative total imports on June 29, 2024, only one day before the End date. Moreover, that quota had been revised to unlimited on April 5, 2024. Under either method, therefore, pineapple never exhausted its quota during No. 3, and the entire designated period appears as black cells in table 1 (treated status). For No. 4 and No. 5, the government announced the quota as unlimited from the outset, leaving no

Table 7: Pineapple Treatment Detail

No	Start date	End date	Quota (ton)
1	Nov. 10, 2022	Dec. 31, 2022	8600
2	Aug. 25, 2023	Dec. 31, 2023	5000
3	Jan. 19, 2024	June 30, 2024	40000 → unlimited
4	July 1, 2024	Sept. 30, 2024	unlimited
5	Oct. 1, 2024	Dec. 31, 2024	unlimited
6	Jan. 24, 2025	June 30, 2025	46000

exhaustion calculation to perform, and both periods are black cells throughout. Finally, No. 6 shows no exceedance under either method and is likewise black throughout.

C Appendix: Tariff Rate Priority Structure

To determine the final applicable rate among multiple tariff rates that may apply to a single HSK code, the priority order of rate application was examined in accordance with *Article 50 of the Customs Act*. The priority order ranges from first to seventh, with the application method as follows.

The rates applied with the highest priority are Special Tariff Rates designed to protect domestic industries or correct trade imbalances, including Anti-Dumping Duty, Retaliatory Duty, and Countervailing Duty. The second-priority rate is the FTA rate under the *Act on Special Cases of the Customs Act for the Implementation of Free Trade Agreements*. If this rate is equal to or lower than rates under the *Customs Act* (excluding first-priority rates), the FTA rate is applied preferentially upon the importer's request. Third-priority rates include International Cooperation Tariffs, such as WTO General Tariff Concessions, and Beneficial Tariffs. Among third-priority rates, Tariff Concessions for agricultural, forestry, and livestock products (conceded at rates equivalent to the domestic-foreign price differential or at rates exceeding Basic Tariff Rates in conjunction with domestic market opening under the *Regulations on Tariff Concessions in the Framework of the Agreement Establishing the World Trade Organization*) take precedence over Basic Tariff Rates and Provisional Tariff Rates. Fourth-priority rates include Adjusted Duty²⁶, Quota Tariff, and Seasonal Tariffs. Quota Tariff takes

²⁶ Article 69, Subparagraphs 1, 3, and 4 of the Customs Act

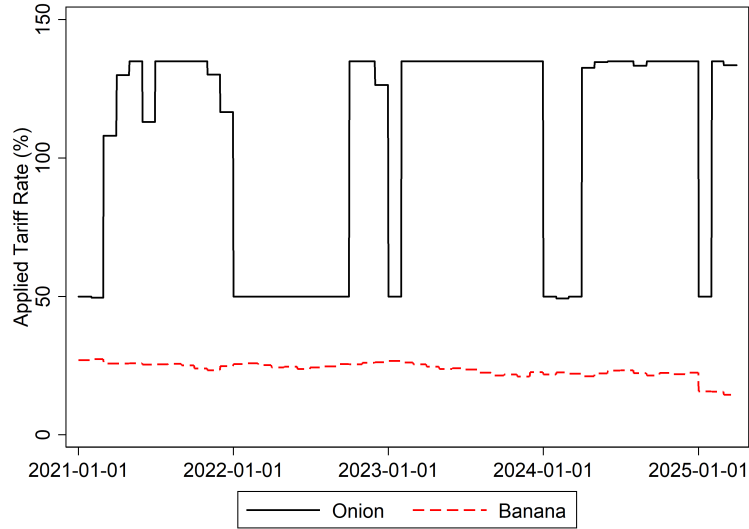
precedence when it is lower than Preferential Tariffs for least developed countries, and it also supersedes Basic Tariff Rates and Provisional Tariff Rates. The fifth-priority rate is Preferential Tariffs for least developed countries. Provisional Tariff Rates and Basic Tariff Rates correspond to the sixth and seventh priority rates, respectively. However, due to data availability limitations, Specific Duties were not considered in this study.

D Appendix: Deriving the Applied Tariff Rate, Worked Examples

For example, as of July 2024, the import shares for banana under HSK code 0803.90-0000 were highest for the Philippines, followed by Vietnam and Ecuador. Ecuadorian banana is not subject to first-priority rates (such as Anti-Dumping Duty, Retaliatory Duty, or Countervailing Duty), and Ecuador is not an FTA partner country; therefore, the review begins with the third-priority rates. Since the fourth- through sixth-priority rates do not apply and the third-priority WTO General Tariff Concession rate (90%) exceeds the seventh-priority Basic Tariff Rate (30%), the Basic Tariff Rate is ultimately applied. Vietnamese banana is subject to a 0% tariff under the Korea-Vietnam FTA, while Philippine banana, which accounts for approximately 61% of imports, is subject to the 30% Basic Tariff Rate. Weighting these country-specific rates by import volume shares yields an applied tariff rate of approximately 23.3% for banana in July 2024. With the Korea-Philippines FTA entering into force on December 31, 2024, the applied tariff rate for banana declined to approximately 14~15% as of the first quarter of 2025, as illustrated in figure 7.

Fresh unpeeled onion is not subject to first-priority rates; thus, the review begins with the second-priority rates. Notably, Chinese onion accounts for the majority of total import volume, but it derives no practical benefit from the FTA Tariff Concessions, and therefore the third-priority rates apply. The third-priority rate is the WTO Tariff Concession applicable to agricultural, forestry, and livestock products with an established Market Access Quota. Imports within the Market Access Quota are subject to the Recommended Tariff Rate of 50%, while imports exceeding the quota are subject to the non-recommended rate of 135%. Consequently, the monthly applied tariff rate for onion varies depending on whether the Market Access Quota is exceeded. This explains the substantial variation, between 50% and 135%, of the applied tariff rate for

Figure 7: Applied Tariff Rate (Banana, Onion)



onion in figure 7. Values that deviate from exactly 50% or 135% reflect onion imports originating from countries other than China.

E Appendix: Removing Seasonality

This paper removes seasonality via a regression-based adjustment employing calendar dummy variables and Fourier terms (Pierce et al., 1984). For daily data, we include day-of-week dummies, model annual seasonality using Fourier (sine/cosine) terms, and control for major and moving holidays with dedicated dummies; the seasonally adjusted series is obtained by subtracting the estimated seasonal components from the raw data.

Two considerations motivated this choice. The first is the need to keep policy effects out of the seasonal component: Quota Tariff designations often coincide with periods of seasonal price fluctuation, and the adjustment must therefore distinguish price movements attributable to the policy from recurring seasonal patterns. A natural candidate for daily deseasonalization is STL, a non-parametric method that extracts seasonality at any specified period and has been applied extensively to weekly and

daily data (Cleveland et al., 1990).²⁷ Since STL relies on non-parametric smoothing, however, trends or policy shocks in the data may be partially absorbed during the decomposition, and treatment effects removed at the seasonal-adjustment stage cannot be recovered in the subsequent DiD analysis.²⁸

The regression-based adjustment eliminates this risk. The calendar dummies and Fourier terms enter as deterministic regressors, independent of the treatment variable, which lets the seasonal component be estimated separately rather than confounded with trends or policy effects. No seasonal pattern remains in the adjusted series, while trend changes and the effects of Quota Tariff designations are fully preserved.

The second consideration is the structure of the raw series itself. The daily data exhibit strong, multi-layered seasonality: week-of-year and day-of-week patterns, holiday effects, and an annual cycle that is not even fixed in length, some years comprising 52 weeks and others 53. Standard official seasonal-adjustment tools such as X-12/X-13-ARIMA-SEATS are not directly applicable to such data, because they are designed for monthly or quarterly series and assume fixed 12-month or 4-quarter cycles (Mollins and Lumb, 2024).

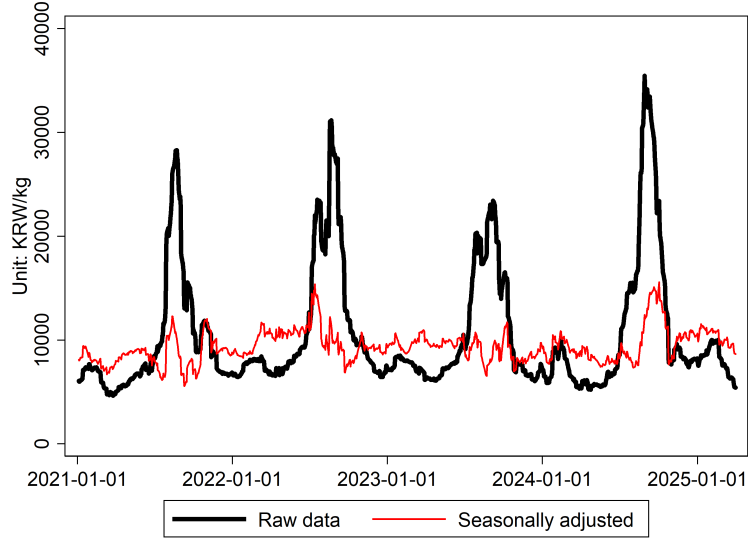
The regression-based approach is established practice for exactly this situation: the U.S. Bureau of Labor Statistics (BLS), when seasonally adjusting weekly unemployment insurance claims, developed and has employed *MoveReg*, a moving weighted regression based on weekly dummies and Fourier terms, instead of traditional X-13 techniques (Mollins and Lumb, 2024).

Figure 8 shows the result for spinach: the adjusted series retains the level and trend of the original while the recurring within-year swings are removed.

²⁷Multiple STL (MSTL) extends the method to decompose several seasonal factors simultaneously, such as 7-day and annual cycles (Bandara et al., 2025).

²⁸STL-based methods also assume fixed seasonal cycles, which leaves them unable to accommodate the week-53 problem or cycle variation attributable to leap years: imposing an annual 52-week cycle presumes that exactly the same cycle repeats every year, which can leave subtle seasonal residuals in year-end weeks. Nor do they offer adjustments for moving holidays or trading-day effects, leaving variation from shifting holiday dates inadequately removed (Mollins and Lumb, 2024).

Figure 8: Seasonally Adjusted Retail Price (Spinach)



F Appendix: Additional Details of the Estimation Design

Clean Treatment Events

Table 8 lists every treatment onset among the ever-treated products after bridging administrative gaps of at most 30 days. For each onset, the table reports spell lengths, the gap since the previous episode, and the clean/dirty classification implied by the 120-day washout condition. Three onsets are dirty: carrot (October 29, 2024; 52-day gap), pineapple (January 19, 2024; 102-day gap), and cabbage (January 24, 2025; 103-day gap). All three are therefore excluded from the treated side in every specification, leaving nine vegetable and eight fruit events.

Treated-Set Attrition in the Persistent-Treatment Specification

Under the stay-on condition, an event remains in the sample that LP-DiD selects for the horizon- h regression only while h is smaller than its spell length, which makes the treated set surviving at each horizon exactly the one recorded by the spell lengths in table 8. The persistent-treatment LP-DiD graphs in the main text are truncated at

Table 8: Quota Tariff Onsets and Washout Classification

Group	Product	Onset date	Spell (days)	Gap (days)	Classification
1	Cabbage	2024-05-10	156	—	clean
1	Cabbage	2025-01-24	67 [†]	103	dirty (washout)
1	Carrot	2024-05-10	120	—	clean
1	Carrot	2024-10-29	154 [†]	52	dirty (washout)
1	Green onion	2022-07-20	104	—	clean
1	Green onion	2023-05-01	61	181	clean
1	Green onion	2023-11-17	166	139	clean
1	Napa cabbage	2024-05-10	326 [†]	—	clean
1	Onion	2022-08-17	177	—	clean
1	Radish	2023-05-01	61	—	clean
1	Radish	2024-07-01	274 [†]	366	clean
2	Avocado	2024-01-19	438 [†]	—	clean
2	Banana	2022-11-10	52	—	clean
2	Banana	2023-11-17	501 [†]	320	clean
2	Kiwifruit	2024-04-05	271	—	clean
2	Mango	2022-11-10	52	—	clean
2	Mango	2023-08-25	585 [†]	236	clean
2	Pineapple	2022-11-10	52	—	clean
2	Pineapple	2023-08-25	45	236	clean
2	Pineapple	2024-01-19	438 [†]	102	dirty (washout)

[†] Spell right-censored at the end of the sample period (March 31, 2025).

Spells are measured after bridging administrative gaps of at most 30 days; gaps are days since the previous episode ended.

$h = 150$ because, on exactly the day of the compositional change at $h = 156$ (the end of the cabbage spell), the coefficient of the vegetable treatment-status specification rises abruptly from $+7.9$ to $+23.7$ log points. This discontinuity is not a natural effect of the Quota Tariff treatment but the effect of a break in the sample composition caused by events failing the stay-on condition. The sharp rise of the vegetable coefficients beyond $h = 156$ must therefore not be interpreted as a causal effect of the Quota Tariff. In the impulse-response specification, there is no stay-on condition in the first place; no event ever exits the sample, and the composition of the treated set is identical at every horizon.

The Washout Window and Long-Horizon Tracking

The washout condition treats a Quota Tariff onset as a clean treated event when the product has never been treated during the preceding $L = 120$ days, under the assumption that the dynamic effects of a previously treated episode stabilize within 120 days (Dube et al., 2025). The impulse-response specification, however, tracks each onset

for up to 273 days from the day it begins, and an episode that ended 120~273 days before a new onset therefore gives rise to the following overlap of episodes. The new episode begins 120~273 days after the previous one and thus satisfies the washout condition, which classifies it as treated. In addition, the past episode, which had begun 120~273 days earlier, also remains classified as treated, because the impulse-response specification requires no stay-on condition in the first place. As a result, the new episode and the past episode remain treated at the same time, and both enter the sample of the LP-DiD regressions. In the data of this study, four Quota Tariff onsets are in this state. Raising L to 273 days would remove this overlap at the cost of discarding clean events; the results of this paper retain $L = 120$. The clean treated-event sets implied by (3) coincide with those produced by the `nonabsorbing()` option of the `lpdid` Stata module (Busch and Girardi, 2024).

G Appendix: Alternatives for Inference with Few Treated Clusters

This appendix works through the alternatives to conventional cluster-robust inference and states why each is unavailable in this setting.

Randomization and placebo procedures (Conley and Taber, 2011; Hagemann, 2019; MacKinnon and Webb, 2020) infer the distribution of the treated clusters' shocks from the control clusters, which is valid only when the two are exchangeable, and MacKinnon and Webb (2018) show that they fail when error variances differ across clusters in unknown form. Exchangeability is untenable here: designation itself selects products prone to price surges, and daily price volatility differs sharply between the domestic vegetables and the imported fruits. Ferman and Pinto (2019) relax exchangeability, but their correction requires the heteroskedasticity to follow a known parametric function of an observable²⁹ by which the control residuals can be re-scaled to stand in for the treated clusters' shocks. The present panel offers no such observable: it is balanced, each product-day observation is a single price rather than a cell average, and the volatility differences follow no known parametric structure. Canay et al. (2021) derive conditions under which the wild bootstrap remains valid with few clusters, but these

²⁹In their leading case, the number of underlying observations averaged into each group $\times$ time cell.

require homogeneity of the covariate design across clusters and are, by the authors' own account, difficult to satisfy when treatment is assigned at the cluster level. Even the recent extensions to staggered timing and heterogeneous effects ([Alvarez and Ferman, 2023](#)) maintain a binary absorbing treatment and a modeled variance structure.